Feature-Driven Priority Queuing

Traditional queuing theory assumes that job types are perfectly observed and assigns each job to a type-specific priority queue—an approach we term *type-driven* priority queuing. We study *feature-driven* priority queuing, where types are unobserved and must be inferred from observable features using a classifier. We examine two implementations. The first, *type-first*, predicts type probabilities from features and then maps these probabilities to priority queues. The second, *direct*, bypasses type prediction and maps features directly to priority queues in an end-to-end manner. The classifiers in both implementations can be trained using labeled data of features (e.g., chest X-rays) with types (e.g., disease findings), but we train the direct classifier to minimize empirical waiting cost rather than type-prediction error.

In the type-first approach, the type classifier is optimized and locked in the first stage; queue assignment is then optimized to minimize an estimated waiting cost computed from the type classifier's output distribution. The actual waiting cost, however, depends on the underlying feature distributions. The estimated waiting cost converges to the actual waiting cost only when the classifier recovers the Bayes posterior—a condition rarely satisfied by complex, misspecified models in high-dimensional settings. The direct approach instead optimizes an empirical waiting cost computed directly from the type-labeled features, ensuring convergence to the actual waiting cost and yielding systematically better queue assignments. We prove this advantage analytically, demonstrate it in tractable examples, and confirm it in large-scale simulations. In experiments with 100,000 chest X-rays and state-of-the-art deep learning classifiers, the direct approach reduces average radiologist waiting cost by 30–60%, driven by its ability to jointly capture delay-cost differences and limited feature separability when assigning priorities.

Key words: data-driven operations, priority queues, digital triage, feature-driven classification

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Priority queues are widely used in resource-constrained settings to differentiate waiting times by job priority and have been extensively studied in the operations literature (e.g., Cobham 1954, van der Zee and Theil 1961, Beja and Sid 1975, Argon and Ziya 2009, Van Mieghem 1995). A growing application is in radiology, where incoming medical images are assigned to priority levels—*routine*, *important*, *urgent*, or *critical*—that determine sequencing in the radiologist's work queue.

Recent advances in artificial intelligence (AI) and machine learning (ML) have enabled classifiers that estimate disease probabilities from features such as shape, texture, density, or anatomical abnormalities in medical images. Such classifiers are increasingly deployed to prioritize cases for radiologists (e.g., Qure AI 2016, DeepTek AI 2017). Motivated by this setting, we study *feature-driven priority queuing*, in which types are unobserved at arrival and must be inferred from observable features. Each job belongs to one of many discrete types, characterized by a feature distribution, service time

distribution, and per-unit delay cost. While the number of types can be large, the number of available queues is typically small—three to five (U.S. Department of Health and Human Services 2025).

When job types are known, the optimal policy—that minimizes the long-run-average waiting cost—assigns jobs to queues based on some version of the celebrated $c\mu$ rule (Cox and Smith 1961), where jobs are prioritized in decreasing order of the product of delay-cost and service rate. In the feature-driven setting, type-conditional feature distributions often overlap and inferring the type from a given set of features is imperfect. Consequently, a feature to queue mapping typically induces a probabilistic mapping from types to queues.

The classification probabilities of types to queues govern queue-specific waiting times and the overall average waiting cost. Moreover, queuing externalities—(i) assigning a job to a queue affects the waiting times of all jobs in its own and lower-priority queues during the busy period, and (ii) a job's waiting time depends on the characteristics of jobs ahead in queue—can lead to counterintuitive outcomes. In particular, a more accurate type classifier may result in a higher average waiting cost, underscoring the need to optimize classifiers for prioritization rather than accurate type classification.

We study two approaches for implementing feature-driven priority queuing. The *type-first* approach follows a modular, plug-and-play design: a classifier first predicts type probabilities from features, and these are mapped to queues to minimize estimated waiting cost given the classifier's output distribution (e.g., Argon and Ziya 2009, Sun et al. 2022b). The *direct* approach bypasses type prediction, instead training a classifier end-to-end to map features directly to queue probabilities so as to minimize the empirical waiting cost.

Our main theoretical result is that the data-driven direct implementation converges almost surely to its theoretical optimum, which—under mild conditions—dominates the theoretical optimum of type-first. In contrast, for the type-first implementation to achieve its theoretical optimum, the type classifier must predict the Bayes posterior almost everywhere—a condition rarely satisfied in high-dimensional or misspecified settings. We present analytical results, tractable examples, and large-scale simulations that demonstrate the direct implementation's advantage.

Beyond stronger performance guarantees, the direct approach with general classifiers, such as neural networks, is also computationally tractable. In the direct approach, empirical gradients asymptotically converge to the theoretical gradients, which ensures the reliability of gradient-based optimization methods. For implementation, we present a batch gradient descent algorithm and establish its convergence to stationary points.

We further derive performance bounds for feature-driven prioritization, showing that high feature overlap—quantified by low total variation distance—restricts type separability and limits cost reduction. In the two-type, two-queue setting, we obtain a closed-form lower bound on average waiting cost as a function of total variation. We also demonstrate that enforcing unbiasedness—i.e., forcing

the relative proportions of jobs across queues to be equal to the proportions of jobs across types—can significantly worsen performance. Because of the nonlinearity of the queuing cost, bias can be helpful.

Using a case of three types and two queues, we show that the optimal queue assignment for the intermediate-cost type depends not only on relative delay-costs but also on the separability of types, captured by pairwise total variation distances. As feature distributions become less distinguishable, "statistical pooling" occurs, and the system effectively makes use of fewer queues. Across all settings, we find that prioritizing high-cost types yields the greatest performance gains, and that the optimal policy is robust to moderate errors in cost parameter estimation.

Finally, we conducted simulations using state-of-the-art deep learning classifiers on 100,000 anonymized chest X-rays from an NIH dataset (Baltruschat et al. 2019). We find that the direct approach reduces average waiting cost by 30%–60% relative to type-first under high utilization. These gains are driven by the queue-aware trade-offs identified in our theory. First, the direct approach allows more aggressive over-classification of the medium delay-cost types ($61\% - 81\%$ join the highest priority queue) than type-first ($14\% - 67\%$ join the highest priority queue). Second, it emphasizes higher accuracy (83%) of correctly routing the highest delay-cost type to the highest priority queue, while accepting a lower accuracy (53%) for routing the lowest delay-cost type to the last queue. In contrast, the type-first approach achieves similar accuracy for both extremes (69% and 71%). Finally, both methods exhibit statistical pooling—using only three of four available queues—a pattern predicted by our theory when feature distributions overlap.

Our contributions are both scholarly and practical. We introduce and characterize an end-to-end direct architecture for feature-driven priority queuing, establish its performance dominance over type-first in realistic settings, and show how it can be implemented with existing classifier architectures. Methodologically, we contribute to the integrated prediction–optimization literature (e.g., Ban and Rudin 2019, Bertsimas and Kallus 2020, Elmachtoub et al. 2025, Elmachtoub and Grigas 2022, Liyanage and Shanthikumar 2005, See and Sim 2010) by introducing a new *queuing loss*—an empirical average waiting cost that is nonlinear in classification probabilities—and showing that it can be minimized using standard ML training methods. Practically, our results demonstrate that classifiers can be optimized directly for service outcomes, leading to significant improvements in domains such as healthcare. These findings are also relevant to the evolving U.S. regulatory guidance: FDA-cleared AI/ML tools have historically been "locked," requiring new premarket review for retraining or modification (U.S. Food and Drug Administration 2019, Benjamens et al. 2020), effectively reinforcing a type-first design. Recent guidance points toward greater flexibility (U.S. Food and Drug Administration 2025, Singh et al. 2025), underscoring the timeliness of our comparison.

1. Literature Review

Our main contribution is a comparison of type-first and direct feature-driven priority queuing. In doing so, we contribute to three broad streams of literature: (1) priority assignment under imperfect information, (2) data-driven learning in queuing systems, and (3) integrated prediction and optimization.

Priority Assignment Under Imperfect Information: Priority queues have been studied extensively since the seminal work of Cobham (1954). Motivated by AI/ML-assisted prioritization in digital radiology, we focus on the setting where job types are unobserved upon arrival and must be inferred from observable features.

van der Zee and Theil (1961) consider a setting in which the type of an arriving job is unobservable for a fraction of customers. For these jobs, a classifier outputs the probability of being the high type, and the optimal policy assigns high, medium, or low priority based on whether this probability lies above, below, or within a threshold interval. Argon and Ziya (2009) also assume a fixed calibrated classifier and show that, under linear delay-costs and two queues, the optimal policy is threshold-based, with the threshold decreasing as the system load increases. Their work highlights how classifier signal variance affects optimal waiting costs. Relatedly, Lee et al. (2024) analyze priority rules under nonlinear convex delay costs and show that prediction errors can be incorporated into a policy that is asymptotically optimal in heavy traffic.

Beja and Sid (1975) study how to assign jobs to N priority queues to minimize average waiting cost when both the delay-cost and the service time are random. When only one of the two is observed, the optimal rule assigns priority based on the observed component and the conditional expectation of the other. Bren and Saghafian (2019), by contrast, assume a job's type is known but its service rate is uncertain, and develop an online learning method to adapt priorities as more data are collected. Sun et al. (2022b) analyze a dynamic triage system with exogenous triage errors and study when triage is beneficial over first in first out, accounting for the delay and misclassification it introduces.

Our work departs from these studies in several ways. First, whereas van der Zee and Theil (1961) and Argon and Ziya (2009) assume a fixed classifier, we endogenize the classifier and optimize it directly for queuing performance. Lee et al. (2024), by contrast, take classification errors as primitives. Our direct approach bypasses type prediction and learns a mapping from features to queues that minimizes average waiting cost. Second, while Beja and Sid (1975) assume that either the delay cost c or the service rate μ is observed, we assume neither is directly observed. Instead, we infer the latent job type—which determines the pair (c, μ)—from observable features via a classifier trained end-to-end. Third, unlike Bren and Saghafian (2019), who treat type as known and focus on estimating service parameters, we assume service parameters are known conditional on type, and focus on inferring the type itself. Finally, relative to Sun et al. (2022b), who model exogenous triage errors, we explicitly

characterize how deviations from the Bayes posterior affect average waiting cost, and we treat the classifier as endogenously trainable.

In addition to these modeling differences, our analysis yields new theoretical insights. We derive performance bounds based on total variation distances between type-conditional feature distributions, showing that smaller distances lead to statistical pooling and less effective queue differentiation. We also show that enforcing an unbiased classifier—one that preserves type proportions across queues—can increase waiting costs, underscoring the importance of optimizing classifiers for queuing objectives. Because of the nonlinearity of the queuing cost, bias can be helpful.

Our findings on statistical pooling relate to prior studies on coarseness in prioritization—e.g., balancing revenue and social welfare (Gurvich et al. 2019) or efficiency-equity trade-offs in resource allocation (Ata et al. 2021). In our setting, the trade-off lies between delay stratification and feature separability across types.

Data-Driven Learning in Queuing Systems: Recent work in data-driven operations explores how learning-based approaches can improve queuing performance in complex service systems. Cohen et al. (2024), Dai and Gluzman (2021), Qu et al. (2022), and Tran et al. (2022) apply reinforcement learning methods to learn control policies in queuing systems that are unknown, hard to model, or intractable. These papers typically assume online or adaptive learning over time. In contrast, our work assumes that type-specific service parameters and queuing dynamics are known and stationary, but job types are unobserved and must be inferred from features using labeled data. Rather than learning queuing policies through interaction, we optimize the mapping from features to queues through batch learning. We contribute new theoretical insights by showing how constraints on feature separability affect optimal queuing performance.

Integrated Prediction and Optimization: In the context of data-driven stochastic optimization, Kleywegt et al. (2002) propose a sample average approximation (SAA) framework for solving stochastic programs using empirical distributions. In the context of stochastic inventory control, Liyanage and Shanthikumar (2005) develop parametric approaches that jointly estimate demand parameters and optimize order quantities. More recent work leverages contextual features: See and Sim (2010) propose robust inventory models that incorporate signals such as oil prices and seasonality; Ban and Rudin (2019) design machine learning algorithms that predict order quantities from features and derive bounds based on feature richness and cost asymmetries. Bertsimas and Kallus (2020) develop a prescriptive prediction framework, where weight functions are estimated from data and used to minimize empirical cost; they prove convergence for empirical objective functions that are decomposable over random variable realizations. Closest to our work, Elmachtoub and Grigas (2022) develop an optimization framework that trains prediction models to directly minimize decision

error in a downstream optimization problem, rather than just prediction error. They establish the asymptotic optimality of their framework, including in settings where the two-step "predict-then-optimize" approach fails due to model misspecification. Relatedly, Elmachtoub et al. (2025) analyze how varying degrees of misspecification affect integrated versus two-step approaches, showing that integration yields performance gains under misspecification.

Our setting introduces structural challenges that differentiate it from prior studies in which the optimization objective is decomposable across realizations of the random variable. Queuing costs are nonlinear and non-decomposable across jobs, and queuing externalities imply that the prioritization of one job influences the delay of others. While the benefits of direct optimization are understood conceptually, the mechanisms through which it improves performance in queuing settings—particularly in the presence of feature overlap and system-wide interactions—have not been formally studied. We show that under classifier misspecification, the type-first approach fails to converge to the optimal solution, whereas our direct approach does. In this sense, our results echo the insights of Elmachtoub and Grigas (2022) and Elmachtoub et al. (2025), but in a queuing context where system-wide externalities make the performance impact of misspecification particularly stark.

Finally, our work contributes to the literature on AI/ML loss functions (Goodfellow et al. 2016) by proposing a new "queuing loss"—an empirical estimate of average waiting cost. Classification to priority queues is conceptually related to the use of ranking loss functions (Cortes and Mohri 2003) in classifier ranking tasks. However, queuing loss depends nonlinearly on the classification probabilities across all jobs in a batch. We show that this loss can be minimized using standard gradient-based optimization methods.

2. Model: T Types and N Priority Queues

We consider the problem of assigning a priority queue to each job (e.g., a patient, customer, or X-ray image) arriving over time at a service facility. Each job is assumed to be of a particular *type*, such as the diagnosis label in the case of medical images. If the facility is busy at the time of arrival, the job may have to wait in a queue. We model this setting as a non-preemptive static priority queue.

Type-i jobs, where $i \in \mathcal{T} = \{1, 2, \ldots, T\}$, arrive according to independent Poisson processes with rates λ_i and have exponentially distributed service times with rates μ_i. This model structure captures key features of many service systems, including radiology workflows where chest X-rays are reviewed sequentially by a radiologist. In both in-house and teleradiology settings, it is common for images from distributed facilities to be routed to a centralized location and processed from a single worklist, making the single-server, non-preemptive, static priority queue a natural abstraction.

We assume service times are independent across types. Denoting the type-i load by $\rho_i = \lambda_i/\mu_i$, the server utilization is $\rho = \sum_{i \in \mathcal{T}} \rho_i$. We assume $\rho < 1$ to guarantee stability. A type-i job incurs a

delay-cost c_i per unit of time waiting for service. The relative values of the delay-costs serve as a proxy for relative urgency. We label types, without loss of generality, so that

$$c_1\mu_1 \geq c_2\mu_2 \geq c_3\mu_3 \geq ... \geq c_T\mu_T.$$

Let N be the number of priority queues, with queue 1 having the highest priority and queue N the lowest. The assignment of jobs to priority queues is evaluated based on its effectiveness in stratifying service speed by priority and reducing waiting times for truly urgent jobs. In this study, the objective of priority assignment is to minimize the long-run waiting cost in queue, averaged over all jobs.

To evaluate waiting costs, we must know the delay-cost rate, which depends on the job type, and the expected waiting time, which depends on the assigned queue. Therefore, the waiting cost depends on how job types are classified into priority queues. We assume the system operates under a stationary regime, and all probabilities and expected values are computed with respect to the steady-state distribution of the queuing system. The overall effect of classification on the arrival rate of jobs to different queues can be captured by a $T \times N$ *classification matrix* P: for all $i \in \mathcal{T}$ and $j \in \mathcal{N}$,

$$P_{ij} = \mathbb{P}(\text{a type-}i \text{ job is assigned to queue } j), \tag{1}$$
$$\textstyle\sum_{j\in\mathcal{N}} P_{ij} = 1.$$

Here, $\mathbb{P}(\cdot)$ denotes probability and P_{ij} is the long run proportion of type i jobs that join queue j.

The arrivals into priority-queue $j \in \mathcal{N}$ follow a Poisson process with total rate

$$\lambda_j(P) = \sum_{i\in\mathcal{T}} \lambda_i P_{ij}.$$

If $1/\mu_i$ denotes the mean service time of type i, then the service time of jobs in queue j is hyper-exponentially distributed with mean

$$\frac{1}{\mu_j(P)} = \sum_{i\in\mathcal{T}} \frac{1}{\mu_i} \cdot \frac{\lambda_i P_{ij}}{\lambda_j(P)} = \frac{\sum_{i\in\mathcal{T}} \rho_i P_{ij}}{\lambda_j(P)},$$

and the delay-cost rate in queue j is

$$c_j(P) = \frac{\sum_{i\in\mathcal{T}} c_i\lambda_i P_{ij}}{\lambda_j(P)} = \frac{\sum_{i\in\mathcal{T}} c_i\mu_i\rho_i P_{ij}}{\lambda_j(P)}.$$

The jobs wait in a N-class $M/G/1$ queue. The stationary waiting time in queue j under classification matrix P, $W_j(P)$, has expectation:

$$\mathbb{E}[W_j(P)] = \frac{\mathbb{E}[\mathcal{R}]}{(1-\sum_{l<j}\rho(l,P))(1-\sum_{l\leq j}\rho(l,P))} \tag{2}$$

where $\mathbb{E}[\mathcal{R}] = \sum_{i\in\mathcal{T}} \lambda_i/\mu_i^2$ is the stationary expected residual service time and

$$\rho(j,P) = \sum_{i\in\mathcal{T}} \rho_i P_{ij} = \sum_{i\in\mathcal{T}} \lambda_i P_{ij}/\mu_i$$

is the net traffic intensity (server utilization) of priority queue $j \in \mathcal{N}$; see Ross (2010, Chapter 8) for a full derivation of average wait times in priority queues.

The mean stationary delay-cost per unit time (the average waiting cost) depends on the classification matrix P through

$$\mathcal{C}(P) = \sum_{i \in \mathcal{T}} \lambda_i c_i (\sum_{j \in \mathcal{N}} P_{ij} \mathbb{E}[W_j(P)]) = \sum_{j \in \mathcal{N}} \mathbb{E}[W_j(P)] (\sum_{i \in \mathcal{T}} c_i \mu_i \rho_i P_{ij}). \tag{3}$$

Using the aggregate queue-level parameters defined above, the following result generalizes the well-established $c\mu$ static priority rule—known to be optimal in multiclass $M/G/1$ settings (Cox and Smith 1961)—to our probabilistic classification setting, where each queue receives a mixture of types.

Proposition 1 (Priority Order of Queues) *Let P denote the classification matrix of jobs of T types into N queues. It is optimal to prioritize queue m over queue n if and only if*

$$c_m(P)\mu_m(P) \geq c_n(P)\mu_n(P).$$

All proofs appear in the Appendix.

Queuing Externalities: The classification matrix P determines how jobs are routed to queues, which in turn determines the average waiting time in each queue and for each job type. These queuing interactions create externalities:

1. The classification of a single job into a queue affects the waiting time of all jobs present (and arriving later) in its own and lower-priority queues during that busy period.
2. The waiting time of a job depends not only on its own characteristics, but also on those of other jobs that are ahead of it—or that may get ahead of it—in the queue.

Because of these externalities, better classification accuracy does not necessarily translate into lower waiting costs, as Example 1 illustrates.

Example 1 (Queuing Externalities: Higher Accuracy Does Not Imply Lower Waiting Cost) *Consider three types ($T = 3$) with delay-costs $c_1 = 25$, $c_2 = 2$, $c_3 = 1$, arrival rates $\lambda_1 = \lambda_2 = \lambda_3 = 1$, and service rates $\mu_1 = \mu_2 = \mu_3 = 3.2$. Suppose there are three priority queues ($N = 3$), and two classifiers with the following classification matrices:*

$$P_1 = \begin{bmatrix} 0.6 & 0.2 & 0.2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad P_2 = \begin{bmatrix} 0.6 & 0.2 & 0.2 \\ 0.25 & 0.50 & 0.25 \\ 0 & 0 & 1 \end{bmatrix}.$$

1. ***Accuracy:*** *P_1 is better—both P_1 and P_2 classify types 1 and 3 correctly with the same accuracy, but P_1 classifies type 2 jobs correctly 100% of the time, compared to 50% under P_2.*

2. ***Waiting Cost:*** *P_2 is better—the average waiting costs under P_1 and P_2 are 75.46 and 69.91, respectively.*

Under P_1, the stationary expected lengths of queues 1, 2, and 3 are 0.22, 0.99, and 12.86, respectively; under P_2, they are 0.34, 0.54, and 13.86. Notably, the 20% of high-cost type 1 jobs that end up in queue 3 face fewer jobs ahead (in queues 1 and 2) under P_2 (0.88) than under P_1 (1.21).

Example 1 suggests that queue assignments optimized directly for waiting costs that account for the downstream impact of misclassifications on all queues can outperform those that focus on predicting job types accurately. This observation is a key motivation behind our study.

Having shown how misclassification induces queuing externalities, we now turn to a benchmark in which type information is perfectly observed.

2.1. Perfect Type Information Benchmark

When types are perfectly observed, queue classification is based on type, resulting in a deterministic binary matrix where each type is assigned to a single queue. Let $I_{T\times N}$ denote such a classification matrix:

$$I_{T\times N}(i,j) = \mathbb{1}\{\text{type } i \text{ is assigned to queue } j\}, \tag{4}$$

and let $\mathcal{C}(I_{T\times N})$ denote the corresponding cost.

Proposition 2 (Optimal Prioritization Under Perfect Type Information) *Given T types such that $c_1\mu_1 > c_2\mu_2 > \ldots > c_T\mu_T$, and any number of $N \le T$ priority queues, let $I^*_{T\times N}$ denote the optimal assignment matrix that minimizes the expected cost:*

$$I^*_{T\times N} = \arg\min_{I_{T\times N}} \mathcal{C}(I_{T\times N}). \tag{5}$$

Then, there exist indices $0 = i_0 < i_1 < \cdots < i_{N-1} < i_N = T$ such that

$$I^*_{T\times N}(i,j) = 1 \quad \textit{if and only if} \quad i_{j-1} < i \le i_j.$$

This result implies that types with higher $c\mu$ values should be always assigned to higher-priority queues. While Beja and Sid (1975) study continuous types, we extend it to discrete types.

3. Feature-Driven Prioritization: Known Feature Distributions

When the true type of a job is not directly observable, it must be inferred from a set of observable *features*. For instance, an X-ray image may be represented by geometric or textural characteristics extracted from the scan—these are the features of the image. We denote the observed feature vector by X, and assume it lies in a p-dimensional feature space $\mathcal{X} \subseteq \mathbb{R}^p$. Feature vectors for type-i jobs are

drawn from a type-specific distribution F_i with density f_i, i.e., $X \sim F_i(\cdot)$ for each $i \in \mathcal{T}$, which in this section we assume are fully known.

The goal in feature-driven prioritization is to use the observed feature vector $X \in \mathcal{X}$ to assign the job to a queue $j \in \mathcal{N} = \{1, 2, \ldots, N\}$. For the remainder of our discussion, we use Δ^M, for a given positive integer M (which in our study corresponds to either the number of types T or the number of queues N), to denote the M-dimensional probability simplex:

$$\Delta^M = \left\{ q \in \mathbb{R}^M : q \geq 0,\ \sum_{j=1}^{M} q_j = 1 \right\}. \tag{6}$$

3.1. Two Approaches: Type-First and Direct

We analyze two implementations of feature-to-queue mapping:

1. **Type Classification and Queue Assignment** ("type-first"): A type classifier $t(\cdot) : \mathcal{X} \to \Delta^T$ maps the p-dimensional feature vector $X \in \mathcal{X}$ to a T-dimensional type-probability vector $z = t(X)$, where $z_i = t_i(X)$ is the predicted probability that the job is of type $i \in \mathcal{T} = \{1, 2, \ldots, T\}$. A second-stage queue assignment $q^t(\beta_t, \cdot) : \Delta^T \to \Delta^N$ maps the T-dimensional type classifier output z to an N-dimensional queue probability vector $q^t(\beta_t, z)$, where β_t denotes the D_t-dimensional parameter vector of the queue mapping. The set of all possible values of β_t, denoted by $\mathcal{B}_t$, is a subset of the D_t-dimensional real space, that is, $\mathcal{B}_t \subseteq \mathbb{R}^{D_t}$, and $q_j^t(\beta_t, z)$ prescribes the probability of assigning output z to queue j.

   Deterministic case: In practice, it may be desirable to route jobs deterministically rather than randomly, ensuring that jobs with the same features are assigned to the same queue. Then we can assign a job to the queue with the highest predicted probability, breaking ties in favor of the higher-priority queue:

$$q_j^{t,\mathrm{det}}(\beta_t, z) = \begin{cases} 1 & \text{if } j = \min\{k : k \in \arg\max_{\ell \in \mathcal{N}} q_\ell^t(\beta_t, z)\}, \\ 0 & \text{otherwise.} \end{cases}$$

2. **Direct Queue Classification** ("direct"): A classifier $q^d(\beta_d, \cdot) : \mathcal{X} \to \Delta^N$ maps each p-dimensional feature vector $X \in \mathcal{X}$ to an N-dimensional queue probability vector $q^d(\beta_d, X)$, where β_d denotes the D-dimensional parameter vector of the queue classifier. The set of all possible values of β_d, denoted by $\mathcal{B}_d$, is a subset of the D-dimensional real space, that is, $\mathcal{B}_d \subseteq \mathbb{R}^D$, and $q_j^d(\beta_d, X)$ prescribes the probability of assigning the job with feature X to queue $j \in \mathcal{N}$.

   Deterministic case: The deterministic variant of the direct method assigns a job to the queue with the highest predicted probability, breaking ties in favor of the higher-priority queue:

$$q_j^{d,\mathrm{det}}(\beta_d, X) = \begin{cases} 1 & \text{if } j = \min\{k : k \in \arg\max_{\ell \in \mathcal{N}} q_\ell^d(\beta_d, X)\}, \\ 0 & \text{otherwise.} \end{cases}$$

Regulatory Motivation and Modeling Implications: The type-first design is widely adopted in regulated domains such as healthcare, where machine learning pipelines are structured to perform type classification first and make downstream decisions based on the predicted type. This sequential design has historically been reinforced by U.S. regulatory practice: FDA-cleared AI/ML-based medical devices have typically been "locked" algorithms, with retraining or modification requiring premarket review (U.S. Food and Drug Administration 2019, Benjamens et al. 2020). With evolving FDA guidance now opening pathways for greater flexibility (U.S. Food and Drug Administration 2025, Singh et al. 2025), it is particularly timely to analyze the implications of this modular design constraint by characterizing the performance differences between the direct and type-first approaches.

3.2. Optimal Type-First and Direct Approaches under Known Feature Distributions

Optimal Type-First Approach: The type-first approach is optimized in two sequential stages.

- **Stage 1: Type Classification:** In Section 4 and Appendix 7.6 we discuss common procedures for training/optimizing type classifiers. We denote the resulting optimal type classifier by $t(\cdot) : \mathcal{X} \to \Delta^T$. (Later we will show that the optimal type classifier in this setting outputs the Bayes' posterior.) At the end of stage 1, the type-classifier is "locked" and cannot be modified.

- **Stage 2: Queue Assignment:** The key feature of type-first implementations is that the queue assignment is optimized solely based on the output distribution, denoted by $b(\cdot)$, of the type classifier locked in stage 1; i.e., stage 2 has no access to the type feature distributions. We mathematize this as follows: Let $z = t(X) \in \Delta^T$ denote the output of the type classifier for a feature vector X, where z_i is the predicted probability of type i. The probability density of classifier outputs, denoted $b(z)$, is induced by $t(\cdot)$ and the marginal feature density $f(X) = \sum_{k=1}^{T} p_k f_k(X)$:

$$b(z) \;=\; \int_{\mathcal{X}} \delta\big(z - t(X)\big)\, f(X)\, dX, \tag{7}$$

where $\delta(\cdot)$ denotes the Dirac delta (Rudin 1987, Folland 1999).

We assume that the family of queue assignment mappings available is $q^t(\beta_t, \cdot)$, where β_t is chosen from the set $\mathcal{B}_t$. Given a β_t, the classification probability of assigning type i to queue j is obtained as the ratio of their joint probability to the marginal probability of type i, computed with respect to the type classifier's outputs as we now explain: Conditioned on the type classifier output z, the probability of type i is z_i, and the probability of assignment to queue j is $q_j^t(\beta_t, z)$. Integrating over the output density $b(z)$ yields the marginal probability of type i and the joint probability of type i with queue j:

$$\mathbb{P}(i) \;=\; \int_{\Delta^T} z_i\, b(z)\, dz, \quad \mathbb{P}(i,j) \;=\; \int_{\Delta^T} z_i\, q_j^t(\beta_t, z)\, b(z)\, dz.$$

The resulting *type-classifier-based* classification probabilities are

$$\widetilde{P}^t_{ij}(\beta_t) = \frac{\mathbb{P}(i,j)}{\mathbb{P}(i)} = \frac{\int_{\Delta^T} z_i\, q^t_j(\beta_t, z)\, b(z)\, dz}{\int_{\Delta^T} z_i\, b(z)\, dz} \quad \forall\, i \in \mathcal{T},\ j \in \mathcal{N}. \tag{8}$$

Using $z = t(X)$ and (7), these can equivalently be written in terms of features as

$$\widetilde{P}^t_{ij}(\beta_t) = \frac{\int_{\mathcal{X}} t_i(X)\, q^t_j(\beta_t, t(X))\, f(X)\, dX}{\int_{\mathcal{X}} t_i(X)\, f(X)\, dX}. \tag{9}$$

Type-First-Optimal: The parameter vector β_t is chosen to minimize the average waiting cost associated with the type-classifier-based classification matrix:

$$\widetilde{\beta}^*_t \in \arg\min_{\beta_t \in \mathcal{B}_t} \mathcal{C}(\widetilde{P}^t(\beta_t)). \tag{10}$$

Performance Loss from Using Type Classifier Outputs Instead of Features. A fundamental limitation of the type-first design is that stage 2 optimizes queue assignment based only on the distribution of classifier outputs, without access to the underlying type-conditional feature densities $\{f_i\}$. This restriction distinguishes type-first from a direct design, where the feature densities are fully available for optimization. To formalize the consequences, we introduce the notion of a *theoretical type-first-optimal* benchmark.

Theoretical Type-First Approach. Suppose stage 2 is allowed to use the feature densities $\{f_i\}$ while retaining the type-first design for queue assignment. Then, for a given β_t, the *theoretical* classification probabilities are

$$P^t_{ij}(\beta_t) = \int_{\mathcal{X}} q^t_j(\beta_t, t(X))\, f_i(X)\, dX. \tag{11}$$

The *Theoretical Type-First-Optimal* parameter vector is

$$\beta^*_t \in \arg\min_{\beta_t \in \mathcal{B}_t} \mathcal{C}(P^t(\beta_t)). \tag{12}$$

Remark 1 (Type-First Optimal May Differ from Theoretical Type-First-Optimal) *If the type-classifier-based classification probabilities differ from their theoretical counterparts, i.e., if $\widetilde{P}^t_{ij}(\beta_t) \neq P^t_{ij}(\beta_t)$ for some i, j, β_t, then the corresponding waiting costs, $\mathcal{C}(\widetilde{P}^t(\beta_t))$ and $\mathcal{C}(P^t(\beta_t))$, may also differ. Consequently, the type-first-optimal solution* (10) *can be strictly suboptimal relative to the theoretical type-first-optimal solution* (12).

Proposition 3 (Bayes Posterior: Type-First-Optimal Equals its Theoretical Optimal) *The type-classifier-based classification probabilities $\widetilde{P}^t_{ij}(\beta_t)$* (9) *in type-first are equal to the theoretical classification probabilities $P^t_{ij}(\beta_t)$* (11) *for all $i \in \mathcal{T}$, $j \in \mathcal{N}$, and $\beta_t \in \mathcal{B}_t$ if and only if*

$$\int_{\mathcal{X}} q^t_j(\beta_t, t(X)) \left[\frac{t_i(X)\, f(X)}{\int t_i(X)\, f(X)\, dX} - f_i(X) \right] dX \;=\; 0 \;\forall\; i \in \mathcal{T},\ j \in \mathcal{N},\ \beta_t \in \mathcal{B}_t. \tag{13}$$

Assuming that

(i) The function class $\{q_j^t(\beta_t, t(X))\}$ *is not orthogonal to any non-zero integrable function* $h(X)$*:*

$$\int_{\mathcal{X}} q_j^t(\beta_t, t(X))\, h(X)\, dX \;=\; 0 \;\forall\; j \in \mathcal{N},\; \beta_t \in \mathcal{B}_t \quad \Longrightarrow \quad h(X) \;=\; 0 \text{ almost everywhere.}$$

and

(ii) The functions $\{f_i(X)\}_{i \in \mathcal{T}}$ *are linearly independent.*

Then equation (13) *holds if and only if* $t_i(X) = \mathbb{P}(i \mid X)$*—the Bayes posterior defined as*

$$\mathbb{P}(i \mid X) \;=\; \frac{p_i f_i(X)}{f(X)} \;=\; \frac{p_i f_i(X)}{\sum_{k=1}^{T} p_k f_k(X)}. \tag{14}$$

In summary, the gap between type-first-optimal and theoretical type-first-optimal arises because stage 2 relies on the classifier's outputs rather than the underlying features for queue assignment. The two coincide only when the classifier outputs the Bayes posterior; otherwise, type-first can be strictly suboptimal.

On Assumptions (i)–(ii) in Proposition 3 Assumption (i) holds when the family of mappings $q^t(\beta_t, t(X))$ is sufficiently expressive. Consider threshold mappings for $T = N = 2$ with $q_1^t(\beta_t, t(X)) = 1$ if $t_1(X) \geq \beta_t$ and $q_1^t(\beta_t, t(X)) = 0$ otherwise, for $\beta_t \in (0,1)$. Suppose $t_1(\cdot)$ is a bijection from $\mathcal{X}$ onto $[0,1]$, and the distribution of $t_1(X)$ admits a density that is positive almost everywhere on $(0,1)$. Then $\int q_1^t(\beta_t, t(X)) h(X)\, dX = 0$ for all $\beta_t \in (0,1)$ implies $h(X) = 0$ almost everywhere. Assumption (ii) is satisfied in common settings where the type-conditional feature distributions are distinct. For example, in the Gaussian location family on $\mathbb{R}^p$ with a common positive definite covariance matrix and distinct means, the densities $\{f_i\}$ are linearly independent.

We now turn to a tractable setting, as in Argon and Ziya (2009), that makes the implications of deviating from the Bayes posterior in type-first transparent. In this setting, we analytically illustrate how such deviations distort the type-classifier-based classification probabilities used for optimizing queue assignment under the type-first approach, leading to systematically suboptimal prioritization. In particular, we explicitly characterize a case in which type-first—with a type classifier that deviates from the Bayes posterior—results in *under-prioritization*, meaning that the proportion of jobs assigned to the high-priority queue is smaller than optimal. This phenomenon is not specific to this example; it recurs in our broader analysis (Section 3.3) and in our radiology case study (Section 5).

Example 2 (Suboptimal Type-First Queue Assignment Under Perturbed Bayes Posterior) *Suppose* $N = T = 2$*, and the type classifier* $t(\cdot)$ *imperfectly predicts the Bayes posterior of a feature* X*: if the classifier output is* $t_1(X) = z$*, then the true value of the Bayes posterior is* $\mathbb{P}(1|X) = z - \epsilon(z)$*, where* $0 \leq z - \epsilon(z) \leq 1$ *for all* $z \in [0,1]$*. Suppose* $z - \epsilon(z)$ *is increasing in* z*; that is, higher classifier*

outputs correspond to higher values of the Bayes posterior. Further suppose that the density $b(z)$ of classifier output z has $b(z) > 0$ for all z in $[0,1]$, and that the classifier is mean-preserving:

$$\int_0^1 zb(z)\,dz = p_1 \quad \text{and} \quad \int_0^1 \epsilon(z)b(z)\,dz = 0.$$

Such a classifier exists (see Figure 1). Further, suppose $\rho < 1$.

- *Under the type-first approach with classifier $t(\cdot)$, Proposition 1 in Argon and Ziya (2009) establishes that the optimal deterministic queue mapping assigns all classifier outputs $z \geq \bar{\tau}$ to queue 1 and $z < \bar{\tau}$ to queue 2, where the threshold $\bar{\tau}$ is the unique solution to*

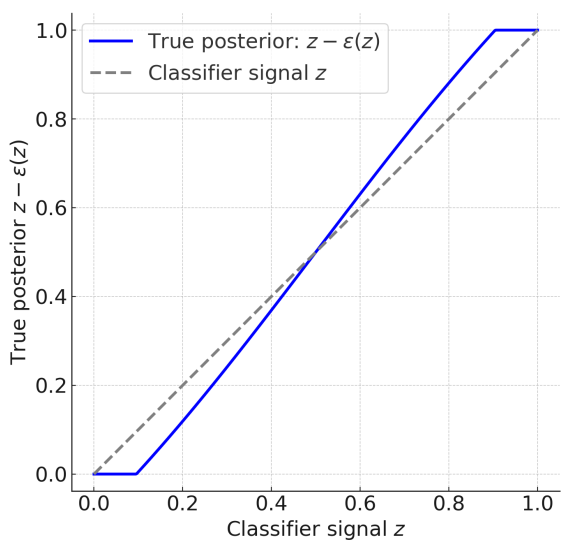
$$\frac{\bar{\tau}}{1-\bar{\tau}} = \frac{p_1 - \rho \int_{\bar{\tau}}^1 zb(z)\,dz}{p_2 - \rho \int_{\bar{\tau}}^1 (1-z)b(z)\,dz}. \tag{15}$$

- *In contrast, the optimal deterministic queue mapping that accounts for the perturbation assigns all outputs $z \geq \tau^*$ to queue 1 and $z < \tau^*$ to queue 2, where τ^* is the unique solution to*

$$\frac{\tau^* - \epsilon(\tau^*)}{1-\tau^* + \epsilon(\tau^*)} = \frac{p_1 - \rho \int_{\tau^*}^1 (z-\epsilon(z))b(z)\,dz}{p_2 - \rho \int_{\tau^*}^1 (1-(z-\epsilon(z)))b(z)\,dz}. \tag{16}$$

Next, we show that if the type classifier flattens the Bayes posterior, as illustrated in Figure 1, it leads to under-triage: fewer jobs are classified into queue 1 than would be optimal.

Figure 1 Illustration of a type classifier flattening the Bayes posterior as in Lemma 1.



Lemma 1 (Flattening the Bayes Posterior Leads to Under-Triage) *Let $\bar{\tau}$ be as in Example 2 (15), and suppose there exists a threshold $\tau_u \leq \bar{\tau}$ such that*

$$0 \leq z - \epsilon(z) \leq z \quad \text{for all } z \leq \tau_u, \quad \text{and} \quad z \leq z - \epsilon(z) \leq 1 \quad \text{for all } z \geq \tau_u.$$

Then $\bar{\tau} \geq \tau^*$, and fewer jobs than optimal are assigned to queue 1 under type-first.

The intuition behind Lemma 1 is that when the classifier overestimates low posterior values ($z - \epsilon(z) \leq z$ for $z \leq \tau_u$), for small classification thresholds, it overestimates the classification probabilities of both types into the first queue. Consequently, it ends up choosing a higher than optimal threshold.

This example highlights that if one has access to features, as in the direct approach which we discuss next, it is possible to correct for the classifier's error profile—even imperfectly—and thereby improve upon the type-first approach, which bases prioritization on the uncorrected classifier signal.

Optimal Direct Approach: We assume that the family of classifiers available is $q^d(\beta_d, \cdot)$, where β_d is chosen from the set $\mathcal{B}_d$. Given a β_d, for all $i \in \mathcal{T}$ and $j \in \mathcal{N}$, the classification probability of type i to queue j is

$$P^d_{ij}(\beta_d) = \int_{\mathcal{X}} q^d_j(\beta_d, X)\, f_i(X)\, dX. \tag{17}$$

Direct-Optimal: the parameter vector β^*_d is chosen to minimize the theoretical average waiting cost:

$$\beta^*_d \in \arg\min_{\beta_d \in \mathcal{B}_d} \mathcal{C}\big(P^d(\beta_d)\big)\,. \tag{18}$$

Why Direct Dominates Type-First. In §4 and §5 we will show that direct dominates type-first. To build some intuition, we propose some of the underlying reasons of this dominance: Direct implementations optimize queue assignment with respect to the full feature distribution, whereas type-first implementations optimize only with respect to the distribution of classifier outputs. In Appendix 7.3, we illustrate in the case $N = T = 2$ how using the same classifier to predict queue probabilities directly can yield a lower waiting cost than when it is used to predict type probabilities in type-first. Moreover, whenever the type classifier maps different feature vectors to the same output, type-first necessarily treats those features identically, while direct can still differentiate among them. In fact, direct can also dominate the theoretical type-first because of its greater structural flexibility: any type-first policy of the form $X \mapsto t(X) \mapsto q^t(\beta_t, t(X))$ can be implemented as a direct policy $X \mapsto q^d(\beta_d, X)$, but the converse may not always hold.

Having argued that the direct approach is structurally more powerful than type-first, we now turn to a complementary question: *what are the fundamental performance limits of feature-driven priority queuing itself?* Even an optimally implemented direct policy cannot overcome the statistical indistinguishability of types when their feature distributions overlap. To capture these limits, we derive lower bounds on the minimum achievable average waiting cost, expressed in terms of total variation distances between type-conditional feature distributions. For clarity, we focus on the practically relevant case of deterministic queue assignment.

3.3. Lower Bound on Achievable Waiting Cost with Feature-Driven Priority Queuing

Suppose the subspace $\mathcal{X}_j$ of feature space $\mathcal{X}$ is assigned to queue $j \in \mathcal{N}$, leading to the classification probabilities:

$$P_{ij} = \int_{\mathcal{X}_j} f_i(X)\, dX \quad \forall i \in \mathcal{T}, j \in \mathcal{N}. \tag{19}$$

Overlapping feature distributions impose constraints on achievable values of P_{ij}. Specifically, for all $i, m \in \mathcal{T}$ and $j \in \mathcal{N}$,

$$|P_{ij} - P_{mj}| \leq TV_{im}, \quad \text{where } TV_{im} = \tfrac{1}{2} \int_{\mathbb{R}^p} |f_i(X) - f_m(X)|\, dX. \tag{20}$$

Here, $TV_{im} = 0$ indicates identical distributions (types indistinguishable from features), while $TV_{im} = 1$ indicates disjoint supports (types "separated" and perfectly distinguishable from features).

Lower Bound on Waiting Cost via Total-Variation Constraints Let P^* denote the optimal classification matrix subject only to total variation constraints, that is,

$$P^* = \arg\min_P \mathcal{C}(P) \quad \text{subject to } |P_{ij} - P_{mj}| \leq TV_{im} \quad \forall i, m, j. \tag{21}$$

Then, $\mathcal{C}(P^*)$ provides a lower bound on the achievable average waiting cost $\mathcal{C}(P)$ over all possible mappings from features to priority queues.

Proposition 4 (Lower Bound on Waiting Cost $N = T = 2$) *For $T = N = 2$, the classification matrix P^* in* (21) *satisfies $P^*_{11} = 1$ and $P^*_{21} = 1 - TV_{12}$. The corresponding average waiting cost is:*

$$\mathcal{C}(P^*) = \mathcal{C}(I^*_{2\times 2}) + \frac{(c_1\mu_1 - c_2\mu_2)\,\mathbb{E}[S]\,\rho_1\rho_2(1 - TV_{12})}{(1-\rho_1)(1-\rho+\rho_2 TV_{12})},$$

where $I^*_{2\times 2}$ denotes the perfect type-information solution defined in (5).

Proposition 4 shows how the lower bound, $\mathcal{C}(P^*)$, on the average waiting cost decreases as the distance between feature distributions (TV_{12}) increases and as ρ_1 or ρ_2 increase. Figure 2 shows how, as the total-variation distance TV_{12} increases, the feasible set in (21) expands and the optimal P^* approaches the perfect type information queue assignment $I^*_{2\times 2}$.

We next examine the achievable waiting cost for an unbiased queue classifier in which the share of jobs sent to each queue is forced to match the overall type proportions, i.e., the proportions of jobs in queues 1 and 2 are p_1 and p_2, respectively. Figure 2 shows how imposing the unbiasedness constraint significantly restricts the set of feasible values of P. By comparing the lowest possible average waiting cost under the unbiasedness constraint with the unconstrained lower bound $\mathcal{C}(P^*)$ derived in Proposition 4, we quantify how much is gained by allowing queues to be deliberately skewed (imbalanced) when features warrant it—flexibility that the direct approach naturally exploits.

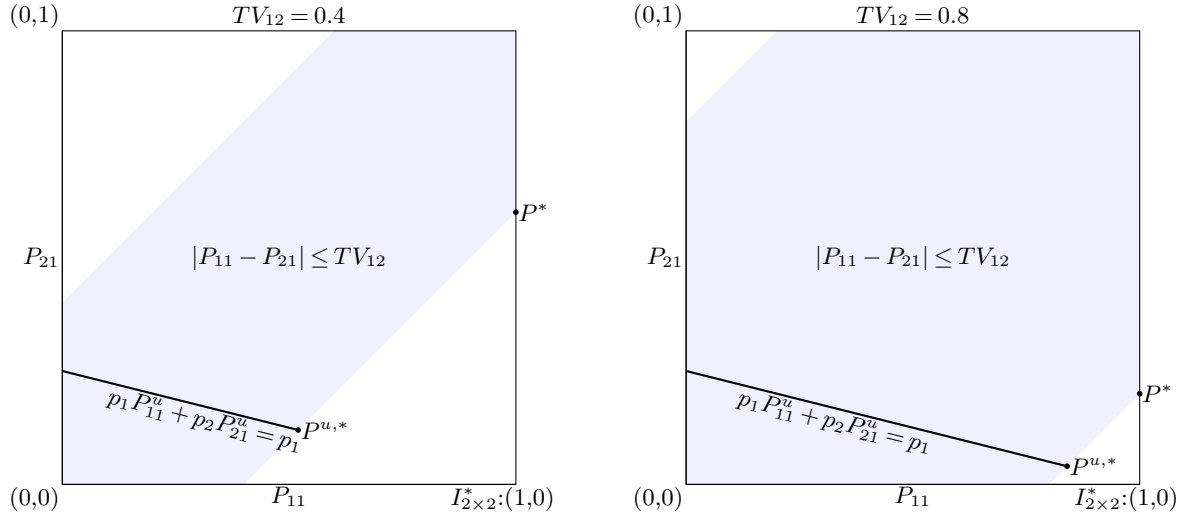


Figure 2 **Feasible classification probabilities P under the total-variation constraints (shaded). The black line marks classification probabilities P^u that also satisfy the unbiasedness constraint $p_1 P^u_{11} + p_2 P^u_{21} = p_1$. P^* and $P^{u,*}$ are the optimal classification probabilities defined in Propositions 4 and 5, respectively. Parameters: $p_1 = 0.2$, $p_2 = 0.8$. Left: $TV_{12} = 0.4$. Right: $TV_{12} = 0.8$.**

Proposition 5 (Lower Bound under Unbiased Classification $N = T = 2$) *Let $P^{u,*}$ denote the optimal unbiased classification matrix, i.e., the matrix with the lowest average waiting cost among all classification matrices P^u that satisfy the unbiasedness constraint $p_1 P^u_{11} + p_2 P^u_{21} = p_1$, in addition to the total variation constraints* (20)*. Then, $P^{u,*}_{11} = p_1 + p_2 TV_{12}$, $P^{u,*}_{21} = p_1(1 - TV_{12})$, and*

$$\mathcal{C}(P^{u,*}) = \mathcal{C}(P^*) + \frac{(c_1\mu_1 - c_2\mu_2)\,\mathbb{E}[S]\,\rho_1\rho_2\rho p_2}{1-\rho} \cdot \frac{TV_{12}(1 - TV_{12})}{(1-\rho+\rho_2 TV_{12})(1-\rho p_1 + (\rho_2 p_1 - \rho_1 p_2)TV_{12})}.$$

Remark 2 (Decrease in Cost from Relaxing Unbiasedness) *Define the relative cost reduction from relaxing unbiasedness as*

$$\Delta = \frac{\mathcal{C}(P^{u,*}) - \mathcal{C}(P^*)}{\mathcal{C}(P^{u,*})}.$$

Then,

$$\Delta = \frac{(\eta - 1)\rho_1\rho_2\rho p_2 TV_{12}(1 - TV_{12}) \times 100}{[\eta\rho_1(1 - \rho p_1 - \rho p_2 TV_{12}) + \rho_2(1 - \rho p_1 + \rho p_1 TV_{12})](1 - \rho + \rho_2 TV_{12})},$$

where

$$\eta = \frac{c_1\mu_1}{c_2\mu_2}.$$

The relative cost reduction (Δ) increases with η, ρ_1, and ρ_2, implying that the benefit of relaxing the unbiasedness requirement is greater when the high-cost job is more costly and the system is more congested. For fixed ρ_1 and ρ_2, the percentage reduction remains bounded above as $\eta \to \infty$.

Together, the unbiasedness and the total variation constraints (20) restrict the feasible values of P_{11}, limiting the extent to which type 1 can be prioritized and thereby possibly increasing the waiting cost. The unbiasedness requirement strictly increases the average waiting cost when feature overlap is intermediate ($0 < TV_{12} < 1$), while the difference vanishes at the extremes ($TV_{12} = 0$ and $TV_{12} = 1$).

3.3.1. Intermediate Types: Lower Bound on Achievable Waiting Cost for $T=3$, $N=2$

In practice, types often form a spectrum—some clearly high or low cost, while others are intermediate. When the number of queues is smaller than the number of types, a natural question arises: should intermediate types be assigned to the high- or low-priority queue?

We focus on the case with three types ($T=3$) and two queues ($N=2$), where type 2 serves as the intermediate type. The summary parameter

$$\gamma := \frac{c_1\mu_1 - c_2\mu_2}{c_2\mu_2 - c_3\mu_3} \tag{22}$$

captures the relative cost-reduction increments; $\gamma=\infty$ implies that distinguishing between types 2 and 3 yields negligible value, while $\gamma=0$ implies types 1 and 2 are nearly indistinguishable in terms of delay-cost. Using γ, the average waiting cost can be expressed as

$$\mathcal{C}(P) = (c_1\mu_1 - c_3\mu_3)\Big[\rho_1 w_1(P) + \frac{\rho_2}{1+\gamma} w_2(P)\Big] + c_3\mu_3 \frac{\rho}{1-\rho},$$

where $w_i(P) = \sum_{j=1}^{N} P_{ij}\mathbb{E}[W_j(P)]$ denotes the expected wait time of type i under P.

Proposition 6 (Optimal Classification of 3 Types into 2 Queues) *For $T=3$ and $N=2$, assume $TV_{12} \le TV_{13},\ TV_{23} \le TV_{13}$, and let P^* be the optimal classification matrix that minimizes the average waiting cost subject to the total variation constraints (21) and the condition $P_{11} \ge P_{21} \ge P_{31}$. Then:*

1. *If $\gamma > \frac{\rho_3(1-\rho+\rho TV_{13})}{\rho_1(1-\rho)}$, then*
$$P^* = \begin{bmatrix} 1, & 0 \\ 1-TV_{12}, & TV_{12} \\ 1-TV_{13}, & TV_{13} \end{bmatrix}$$
2. *If $\gamma \in \left[\max\left\{0, \frac{\rho\rho_2\rho_3 TV_{23} - \rho_3\rho_1(1-\rho)}{(\rho_2+\rho_3)\rho_1(1-\rho)}\right\}, \frac{\rho_3(1-\rho+\rho TV_{13})}{\rho_1(1-\rho)}\right]$, then*
$$P^* = \begin{bmatrix} 1, & 0 \\ 1-TV_{13}+TV_{23}, & TV_{13}-TV_{23} \\ 1-TV_{13}, & TV_{13} \end{bmatrix}$$
3. *If $\gamma < \max\left\{0, \frac{\rho\rho_2\rho_3 TV_{23} - \rho_3\rho_1(1-\rho)}{(\rho_2+\rho_3)\rho_1(1-\rho)}\right\}$, then*
$$P^* = \begin{bmatrix} 1, & 0 \\ 1, & 0 \\ 1-TV_{23}, & TV_{23} \end{bmatrix}$$

Here, the assumptions $TV_{12} \le TV_{13},\ TV_{23} \le TV_{13}$ mean that the feature distributions somewhat preserve the rank ordering of the three types. Type 1 is closer in $c\mu$ to type 2 than to type 3, and the assumptions preserve that closeness in terms of feature distributions as well.

From the different possible values for the optimal classification matrices, we conclude that

Prioritizing High-Cost Types is More Important: The classifier favors protecting type 1 even if it means allowing type-3 into the high-priority queue.

Over/Under-Prioritization of Intermediate Type: A large γ leads to under-prioritizing type 2 to shield type 1. When γ is small, type 2 is over-prioritized at the expense of type-3. As γ decreases further, type 2 merges with type 1.

Statistical Pooling: As total variation distances shrink, types become indistinguishable and are pooled into a single queue.

Robustness: Optimal classification matrices are piecewise constant in γ. Consequently, small errors in estimating delay costs or service rates—and thus in γ—are unlikely to change the optimal queue assignments, except near boundary values of γ. Even when a boundary is crossed, a misestimated γ may select a different classification matrix, but the resulting increase in average waiting cost is small.

In §5, we illustrate these results in a case study on chest X-rays with $T = 14$ types and $N = 4$ queues.

In this section, we analyzed two implementations for feature-driven priority queuing—type-first and direct—and highlighted the structural limitations of the former. We also derived lower bounds on the achievable average waiting cost in terms of total variation distances between type-conditional feature distributions, thereby underscoring the inherent performance limits imposed by overlapping feature distributions. Our analysis was for an idealized setting of known feature distributions. In practice, however, feature distributions are not a priori known, and any implementation of feature-driven priority queuing must rely on data for optimization.

4. Feature-Driven Prioritization: Data-Driven Implementation

This section analyzes our two implementations of feature-driven priority queuing using an observed, labeled dataset $\mathcal{H}_S = \{(X_s, y_s) : s \in \{1, 2, \ldots, S\}\}$ of S i.i.d. samples, where each X_s is a feature vector and $y_s \in \{0,1\}^T$ is a $T-$dimensional type vector such that

$$y_{is} = \begin{cases} 1 & \text{if the type of } X_s \text{ is } i, \\ 0 & \text{otherwise.} \end{cases}.$$

***Direct**:* The data-driven implementation of the direct approach involves optimizing the queue classifier parameter vector β_d to minimize the empirical estimate of the average waiting cost based on the dataset $\mathcal{H}_S$. Specifically, we estimate the optimal parameter vector by solving:

$$\widehat{\beta}_d^* = \arg\min_{\beta_d} \mathcal{C}(\widehat{P}^d(\beta_d)), \tag{23}$$

where $\widehat{P}_{ij}^d(\beta_d)$ are empirical estimates of the classification probabilities $P_{ij}^d(\beta_d)$ (17): $\forall\ i \in \mathcal{T},\ j \in \mathcal{N}$,

$$\widehat{P}_{ij}^d(\beta_d) = \frac{\sum_{s=1}^S y_{is}\, q_j^d(\beta_d, X_s)}{\sum_{s=1}^S y_{is}}. \tag{24}$$

Type-First: The type-first approach optimizes queuing in two sequential stages:

- **Stage 1: Optimizing the Type Classifier:** In Appendix §7.6, we summarize common parametric and non-parametric models—such as maximum likelihood estimation (MLE), Bayesian estimation, method of moments (MoM), and k-Nearest Neighbors (k-NN)—used to estimate the type posterior $t(X)$ from the dataset $\mathcal{H}_S$. These methods have been studied extensively in the literature (Greene 2018, Millar 2011, Gelman et al. 2013, Bernardo and Smith 2009). This step produces the type classifier $t(\cdot)$, which maps $\mathcal{H}_S = \{(X_s, y_s) : s = 1, \ldots, S\}$ to predicted type posteriors $\{t(X_s)\}$. The type classifier's outputs form the sole input for the second step, where their empirical distribution is used to estimate the classification matrix $\widehat{P}^t(\beta_t)$ and the corresponding average waiting cost $\mathcal{C}(\widehat{P}^t(\beta_t))$ for any choice of queue assignment parameter β_t.
- ***Stage 2: Optimizing the Queue Assignment:*** Given the predicted type posteriors from Step 1, the type-first approach maps each $t(X)$ to a queue assignment probability $q_j^t(\beta_t, t(X))$ for each queue $j \in \mathcal{N}$. The queue mapping parameters β_t are then optimized to minimize the estimated average waiting cost based on the type classifier's outputs $\{t(X_s)\}$ from the previous step. Specifically, we solve:

$$\widehat{\beta}_t^* = \arg\min_{\beta_t} \mathcal{C}(\widehat{P}^t(\beta_t)), \tag{25}$$

where $\widehat{P}_{ij}^t(\beta_t)$ are the empirical estimates of the type-classifier-based classification probabilities $\widetilde{P}^t(\beta_t)$ (9), computed using the type classifier's outputs: $\forall\ i \in \mathcal{T}, j \in \mathcal{N}$,

$$\widehat{P}_{ij}^t(\beta_t) = \frac{\sum_{s=1}^S t_i(X_s)\, q_j^t(\beta_t, t(X_s))}{\sum_{s=1}^S t_i(X_s)}. \tag{26}$$

To illustrate the data-driven implementation of the direct and type-first approaches and compare their performance, we now present a stylized example in which the optimal policy under known feature distributions is analytically tractable.

Example 3 (Direct vs. Type-First for a One-Dimensional Feature) *We compare the direct and type-first approaches in a stylized setting with two types ($T = 2$) and two queues ($N = 2$). Type-1 is rare but costly ($p_1 = 0.1$, $c_1 > c_2$); both types have equal service rates ($\mu_1 = \mu_2$), and the system's total utilization is $\rho = 0.7$. The type of a job is unobserved at arrival and must be inferred from a one-dimensional feature X, which follows a uniform distribution on $[0, \frac{3}{2}]$ for type 1 and $[-1, 1]$ for type 2. The optimal policy under known feature distributions assigns all jobs with $X \geq 0$ to queue 1, yielding optimal classification probabilities: $P_{11}^{opt} = 1$, $P_{12}^{opt} = 0$, $P_{21}^{opt} = 0.5$, and $P_{22}^{opt} = 0.5$.*

Empirical Setup. We generate a labeled dataset of $S = 10{,}000$ samples by first drawing the type of each job according to the population distribution ($p_1 = 0.1$, $p_2 = 0.9$). Conditional on the type, we sample the feature X from the corresponding distribution. We use this training dataset to estimate the

direct and type-first classifiers, and evaluate their performance on an independent test set generated in the same way.

- **Direct:** We use a logistic queue classifier $q^d(\beta_d, X)$ that predicts the probability of assigning feature X to queue 1 and queue 2 as $q_1^d((\beta_{d0}, \beta_{d1}), X)$ and $1 - q_1^d((\beta_{d0}, \beta_{d1}), X)$, respectively, where

$$q_1^d((\beta_{d0}, \beta_{d1}), X) = \frac{\exp(\beta_{d0} + \beta_{d1} X)}{1 + \exp(\beta_{d0} + \beta_{d1} X)},$$

and β_{d0} and β_{d1} are optimized to minimize the empirical waiting cost, as per equations (23) and (24). Each feature X is then deterministically assigned to the queue with the higher predicted probability.

- **Type-First:** We implement the two-stage approach with various type classifiers described in Appendix §7.6 (see Appendix §7.7 for their implementation in this example), including: the MLE with a logistic model of the type posterior, Bayesian estimation with Gaussian feature distributions, Bayesian estimation with uniform feature distributions, the MoM with Gaussian features, the MoM with uniform features, and k-NN (with $k = 10$ and $k = 100$). Each classifier's output $t(X)$ is passed to a logistic queue assignment function $q^t(\beta_t, t(X))$, where

$$q_1^t((\beta_{t0}, \beta_{t1}), t(X)) = \frac{\exp(\beta_{t0} + \beta_{t1} t(X))}{1 + \exp(\beta_{t0} + \beta_{t1} t(X))},$$

and β_{t0} and β_{t1} are optimized to minimize the estimated waiting cost based on the type classifier's outputs, as described in equations (25) and (26). Each type classifier output $t(X)$ is then deterministically assigned to the queue with the higher probability.

Table 1 Classification Probabilities and Average Wait Times under Direct and Type-First.

Approach	P_{11}	P_{21}	Type 1 Wait	Type 2 Wait	**Approach**	P_{11}	P_{21}	Type 1 Wait	Type 2 Wait
Theoretical Optimum	1.00	0.50	0.79	1.73	**Direct**	1.00	0.50	0.79	1.73
Type-First					**Type-First**				
MLE Logistic	0.88	0.41	0.92	1.71					
Bayesian (Gaussian)	0.91	0.43	0.89	1.72	Bayesian (Uniform)	1.00	0.49	0.79	1.73
MoM (Gaussian)	0.91	0.43	0.89	1.72	MoM (Uniform)	0.99	0.49	0.81	1.73
k-NN (k=10)	0.84	0.37	0.94	1.71	k-NN (k=100)	1.00	0.50	0.79	1.73

Results. Table 1 reports the classification probabilities and average wait times on the test dataset. The data-driven direct approach exactly recovers the optimal queue assignment under known feature distributions. In contrast, type-first achieves the theoretical optimal performance only when the type classifier closely approximates the Bayes posterior—such as under correctly specified parametric models or non-parametric k-NN with sufficiently large k. We now formally establish the asymptotic convergence properties of the data-driven implementations of the direct and type-first approaches.

Proposition 7 (Data-Driven Direct Converges to its Theoretical Optimum) *Under the direct approach, for any $\beta_d \in \mathcal{B}_d$, the empirical classification probabilities $\widehat{P}^d_{ij}(\beta_d)$ defined in* (24) *converge almost surely to their theoretical values $P^d_{ij}(\beta_d)$ in* (17)*, i.e., as $S \to \infty$, for all $i \in \mathcal{T}$, $j \in \mathcal{N}$,*

$$\widehat{P}^d_{ij}(\beta_d) \xrightarrow{a.s.} P^d_{ij}(\beta_d), \quad i.e., \quad \mathbb{P}\left(\lim_{S\to\infty} \widehat{P}^d_{ij}(\beta_d) = P^d_{ij}(\beta_d)\right) = 1.$$

Moreover, if the parameter space $\mathcal{B}_d$ is compact and $q^d_j(\beta_d, X)$ is continuous in β_d for each X, then

$$\min_{\beta_d \in \mathcal{B}_d} \mathcal{C}\big(\widehat{P}^d(\beta_d)\big) \xrightarrow{a.s.} \min_{\beta_d \in \mathcal{B}_d} \mathcal{C}\big(P^d(\beta_d)\big),$$

and for any empirical minimizer $\widehat{\beta}_d \in \arg\min_{\beta_d} \mathcal{C}(\widehat{P}^d(\beta_d))$, $\mathcal{C}\big(P^d(\widehat{\beta}_d)\big) \xrightarrow{a.s.} \min_{\beta_d} \mathcal{C}\big(P^d(\beta_d)\big)$.

Proposition 7 establishes that, as the sample size $S \to \infty$, the minimum empirical waiting cost converges to the minimum theoretical waiting cost, and the theoretical waiting cost under a minimizer of the empirical waiting cost converges to the minimum theoretical waiting cost. Thus, the data-driven direct approach asymptotically recovers the performance of the theoretical direct approach under known feature distributions discussed in § 3.2. In contrast, this does not always happen with type-first.

Remark 3 (Data-Driven Type-First Converges, But Not To its Theoretical Optimum) *Under the type-first approach, for any queue assignment parameter vector β_t, the empirical estimates of the classification probabilities $\widehat{P}^t_{ij}(\beta_t)$* (26) *converge almost surely to the type-classifier-based classification probabilities $\widetilde{P}^t_{ij}(\beta_t)$* (9)*: for all $i \in \mathcal{T}$, $j \in \mathcal{N}$,*

$$\widehat{P}^t_{ij}(\beta_t) \xrightarrow{a.s.} \widetilde{P}^t_{ij}(\beta_t), \quad i.e., \quad \mathbb{P}\left(\lim_{S\to\infty} \widehat{P}^t_{ij}(\beta_t) = \widetilde{P}^t_{ij}(\beta_t)\right) = 1.$$

However, from Proposition 3, these type-classifier-based probabilities $\widetilde{P}^t_{ij}(\beta_t)$ may not coincide with the theoretical classification probabilities $P^t_{ij}(\beta_t)$ (11) *unless the type classifier predicts the Bayes posterior almost everywhere. Failing that, the empirical waiting cost $\mathcal{C}(\widehat{P}^t(\beta_t))$ may not converge to the theoretical waiting cost $\mathcal{C}(P^t(\beta_t))$, and the minimizers of the former may not minimize the latter.*

In practice, perfectly predicting the Bayes posterior is not straightforward—standard type classifiers rely on strong conditions such as correct parametric model specification, injective moment maps, or posterior continuity. These assumptions are difficult to satisfy in high-dimensional or complex settings, where model misspecification, data sparsity, or the curse of dimensionality can severely degrade performance (see Appendix §7.6 for details).

4.1. Data-Driven Direct Implementation: Convergence of Gradient-Based Methods

Here we show how the direct approach, namely the minimization of the empirical waiting cost $\mathcal{C}(\widehat{P}^d(\beta_d))$, can be implemented using gradient-based algorithms, which are widely used for training nonlinear classifiers such as neural networks (Goodfellow et al. 2016). Such methods typically converge to stationary points of the empirical cost, which are the natural candidates for minimizers.

We show that under mild conditions, the gradients of the empirical waiting cost converge to the gradients of the theoretical waiting cost, and the theoretical gradient evaluated at an empirical stationary point converges to 0. Thus, empirical stationary points asymptotically behave like stationary points of the theoretical waiting cost. Let $\beta_d = (\beta_{d1}, \ldots, \beta_{dD}) \in \mathcal{B}_d \subseteq \mathbb{R}^D$, where $\mathcal{B}_d$ is compact.

The convergence results below follow from standard optimization theory—specifically, uniform laws of large numbers (Jennrich 1969) and gradient descent convergence (Nemirovski 2009, Bottou et al. 2018). What we must verify in our proofs is that the queuing loss $C(\widehat{P}^d(\beta_d))$, which is nonlinear and non-decomposable across samples, satisfies the smoothness conditions required by the general theory.

Proposition 8 (Direct: Empirical Gradient Converges to Theoretical Gradient) *Suppose that the partial derivatives of $q_j^d(\beta_d, X)$ with respect to $\beta_{d1}, \ldots, \beta_{dD}$ exist, are continuous in β_d, and are uniformly bounded over $\beta_d \in \mathcal{B}_d$ and $X \in \mathcal{X}$. Then, for any $\beta_d \in \mathcal{B}_d$, as $S \to \infty$,*

$$\nabla_{\beta_d} \mathcal{C}(\widehat{P}^d(\beta_d)) \xrightarrow{a.s.} \nabla_{\beta_d} \mathcal{C}(P^d(\beta_d)).$$

In particular, if $\widehat{\beta}_d$ is an empirical stationary point with $\nabla_{\beta_d} \mathcal{C}(\widehat{P}^d(\widehat{\beta}_d)) = 0$, then $\nabla_{\beta_d} \mathcal{C}(P^d(\widehat{\beta}_d)) \xrightarrow{a.s.} 0$.

Next, we establish that batch gradient descent converges to stationary points of the empirical waiting cost. For tractability, we take $N = 2$ but the proof justifies that the result extends to $N \geq 3$.

Empirical Waiting Cost ($N = 2$). Consider a two-queue setting with a direct classifier that assigns each feature vector X to queue 1 with probability $q_1^d(\beta_d, X)$ and to queue 2 with probability $1 - q_1^d(\beta_d, X)$. Given a labeled dataset $\mathcal{H}_S$, the empirical waiting cost $\mathcal{C}(\widehat{P}^d(\beta_d))$ using the classification matrix $\widehat{P}^d(\beta_d)$ in (24) can be expressed as follows (see Appendix §7.4 for a full derivation):

$$\mathcal{C}(\widehat{P}^d(\beta_d)) = \frac{\mathbb{E}[\mathcal{R}]}{(1-\rho)} \left[\frac{K - \rho \sum_{s=1}^{S} \eta_s q_1^d(\beta_d, X_s)}{1 - \sum_{s=1}^{S} \gamma_s q_1^d(\beta_d, X_s)} \right], \tag{27}$$

where

$$K = \sum_{i=1}^{T} c_i \mu_i \rho_i, \quad S_i = \sum_{s=1}^{S} y_{is}, \quad \eta_s = \sum_{i=1}^{T} \frac{c_i \mu_i \rho_i y_{is}}{S_i}, \quad \gamma_s = \sum_{i=1}^{T} \frac{\rho_i y_{is}}{S_i}, \quad \mathbb{E}[\mathcal{R}] = \sum_{i=1}^{T} \frac{\lambda_i}{\mu_i^2}.$$

Batch Gradient Descent for Stationary Points of Empirical Waiting Cost

Input: Dataset $\mathcal{H}_S$, initial parameters β_d^0, iterations M **Initialize:** $m \leftarrow 0$

1. **Repeat until** $m = M$**:**
   (a) Compute gradient:
$$\nabla_{\beta_d}\mathcal{C}(\widehat{P}^d(\beta_d^m)) = \mathcal{C}(\widehat{P}^d(\beta_d^m))\Big[\frac{\sum_{s=1}^S \gamma_s \nabla_{\beta_d} q_1^d(\beta_d^m, X_s)}{1-\sum_{s=1}^S \gamma_s q_1^d(\beta_d^m, X_s)} - \frac{\rho}{K}\cdot\frac{\sum_{s=1}^S \eta_s \nabla_{\beta_d} q_1^d(\beta_d^m, X_s)}{1-\frac{\rho}{K}\sum_{s=1}^S \eta_s q_1^d(\beta_d^m, X_s)}\Big].$$
   (b) Choose step size: $\alpha_m = \arg\min_{\alpha\geq 0} \mathcal{C}\big(\widehat{P}^d(\beta_d^m - \alpha\nabla_{\beta_d}\mathcal{C}(\widehat{P}^d(\beta_d^m)))\big)$.
   (c) Update parameters: $\beta_d^{m+1} \leftarrow \beta_d^m - \alpha_m \nabla_{\beta_d}\mathcal{C}(\widehat{P}^d(\beta_d^m))$.
   (d) Increment $m \leftarrow m+1$.
2. **Return:** β_d^M.

Lemma 2 (Direct: Batch Gradient Descent Converges to Stationary Points) *Suppose the first and second partial derivatives of $q_1^d(\beta_d, X)$ with respect to $\beta_{d1}, \ldots, \beta_{dD}$ exist and are uniformly bounded over $\beta_d \in \mathcal{B}_d$ and $X \in \mathcal{X}$. Then, for any initialization β_d^0, batch gradient descent applied to $\mathcal{C}(\widehat{P}^d(\beta_d))$ converges to a stationary point; in particular,*

$$\min_{0\leq m<M} \left\|\nabla_{\beta_d}\mathcal{C}(\widehat{P}^d(\beta_d^m))\right\|_2^2 \to 0 \quad \textit{as } M \to \infty.$$

Lemma 2 shows that the squared gradient norm converges to zero as $M \to \infty$, ensuring convergence to a stationary point of $\mathcal{C}(\widehat{P}^d(\beta_d))$ from any initialization. In the proof, we justify why the result extends to $N \geq 3$ in general. Appendix §7.5 discusses practical considerations for implementation.

For analytical clarity, we focus on batch gradient descent, which computes the gradient using the full dataset. In practice, our case study in §5 employs stochastic gradient descent (SGD), a more scalable method that uses minibatches. Convergence of SGD under similar smoothness assumptions is well established (Bottou et al. 2018), and our experiments confirm its effectiveness.

While we do not provide a full sensitivity analysis of the optimal direct solution, Appendix §7.4 shows that for a linear classifier, minimizing $\mathcal{C}(\widehat{P}^d(\beta_d))$ (27) reduces to a linear program, for which standard sensitivity results apply (Murty 1983). In our case study (§5), the optimal direct classifier implemented with a convolutional neural network is also robust to small parameter changes.

5. Quantifying Improvements using a Chest X-ray 2D Image Dataset

We use state-of-the-art 2D image classifiers to evaluate feature-driven prioritization with $T = 14$ types and $N = 4$ queues, estimating the gains from the direct approach over the type-first approach and identifying their key drivers.

Table 2 **Rank order of delay-costs of diseases and type (highest rank of disease present).**

Disease	Rank Order of Delay-Cost	Image Type (Highest Rank)	Arrival Rate (Prevalence)
Pneumothorax	1	1	0.05
Emphysema	2	2	0.02
Pneumonia	3	3	0.01
Edema	4	4	0.02
Consolidation	5	5	0.04
Effusion	6	6	0.09
Infiltration	7	7	0.12
Atelectasis	8	8	0.04
Cardiomegaly	9	9	0.01
Pleural Thickening	10	10	0.01
Fibrosis	11	11	0.01
Mass	12	12	0.02
Nodule	13	13	0.02
No Finding	14	14	0.54
		Total	**1.00**

5.1. Simulation Study Design

We simulate a single-server priority queuing system to triage chest X-rays (CXRs) at a radiology center using MobileNet-based image classification. We compare the two queue assignment approaches—type-first and direct— and benchmark them against the best and the worst performance algorithms—perfect type information and single queue first-in-first-out, respectively—across multiple delay-cost functions and high total utilization ($\rho = 0.9$).

Data: The publicly available "ChestX-ray14" dataset from the National Institutes of Health (NIH) Clinical Center (Wang et al. 2017, Baltruschat et al. 2019) contains 112,120 frontal chest X-ray (CXR) 2D images with disease labels extracted from corresponding radiological reports, spanning 30,805 unique patients.

Features: Following standard practice, we use an image resolution of 512×512 pixels with 3 RGB (red, green, blue) channels. Each image is represented as a tensor in the feature space $\mathcal{X} \subseteq \mathbb{R}^{512\times512\times3}$.

Types: There are 13 disease findings, which may co-occur or be absent altogether ("no finding"). Table 2 shows the rank order of delay-costs associated with these diseases, based on discussions with clinicians. While this ordering is subjective and context dependent—for example, the urgency of a finding may change during an outbreak, or a repeated CXR showing the same abnormality may be less urgent—the ordering provides a consistent basis for prioritization. We define the type of an image as the rank of the most urgent (i.e., highest delay-cost) disease present. This results in 14 image types, where type 14 corresponds to no disease finding.

Delay-Costs: We consider two families of delay-cost functions:

- *Delay-costs increase in a convex manner across types*: The delay-cost for type i is given by $c_i = \eta^{T-i}$, where $\eta > 1$ governs the degree of cost heterogeneity. This formulation ensures that higher-ranked types incur increasingly higher penalties. For example, $\eta = 1.8$ implies sharper cost separation than $\eta = 1.5$, with differences between successive types satisfying: $c_1 - c_2 > c_2 - c_3 > \ldots > c_{13} - c_{14}$.
- *Delay-costs increase in a linear manner across types*: The delay-cost for type i is $c_i = \zeta(T-i)+1$, where ζ is a constant. This results in uniform cost increments across types: $c_1 - c_2 = c_2 - c_3 = \ldots = c_{13} - c_{14} = \zeta$. We experiment with $\zeta \in \{1.8, 10\}$, where larger ζ values encode stronger prioritization.

Delay-Costs as Design Parameters: Prior work has often treated delay costs either as fixed model primitives (e.g., Cobham 1954, Cox and Smith 1961, Argon and Ziya 2009) or as quantities to be estimated empirically from observed behavior (Aksin et al. 2013). A practical alternative, explored in settings with service differentiation, is to view delay costs as design parameters chosen to induce desirable system performance (e.g., Celik and Maglaras 2008, Maglaras et al. 2018). In our setting, one can adopt the latter perspective by treating delay costs as design parameters chosen to induce desirable waiting-time profiles across types. Different delay-cost structures yield different optimal assignments $I^*_{T\times N}$ of types to queues under perfect type information, which in turn induce different waiting-time profiles across types. One can therefore choose a cost structure that—under perfect type information—produces the waiting-time profile that a radiology center finds suitable. For illustration, the "Perfect" column in Table 4 in the Appendix reports the waiting times for different types under perfect type information for various cost structures. Choosing delay costs that increase linearly across types leads to the first five types having the same waiting time as type 1 (0.93), while type 14 alone incurs a long wait (13.82). In contrast, choosing delay costs that increase convexly across types leads to the first three types having the same waiting time as type 1 (0.87), while types 13 and 14 both incur long waits (13.41).

Arrival Rates: For the purposes of this study, we assume that arrivals of CXRs of different types to the radiology center follow independent Poisson processes. We normalize the total arrival rate to 1, so that the arrival rate of each type equals its frequency in the full dataset of images. For the dataset $\mathcal{H}_S = \{(X_s, y_s) : s \in \{1, 2, \ldots, S\}\}$, where each sample consists of a feature array X_s (a $512 \times 512 \times 3$ array with red, green, and blue pixel intensities) and a type label y_s (see §4 for a detailed description), we estimate the arrival rate of type-$i \in \mathcal{T}$ images as $\hat{\lambda}_i = \sum_{s=1}^{S} t_{i,s}/S$. Table 2 reports the estimated arrival rates for each image type.

Service Rates: We assume identical service time distributions of the radiologist for all image types: $\mu = (\sum_{i\in\mathcal{T}} \hat{\lambda}_i)/\rho \ \forall i \in \mathcal{T}$, where ρ is the total server utilization.

Queues: In large radiology centers and hospitals, it is common practice to assign each incoming CXR a priority queue for radiologist review (a process often referred to as "triage"). Typically, there are four priority queues: *Critical* (1) $\succ$ *Urgent* (2) $\succ$ *Important* (3) $\succ$ *Routine* (4).

Image Classifiers: Existing studies (Wang et al. 2017, Baltruschat et al. 2019) and open-source implementations (Garyfallos et al. 2019) use deep learning methods for image classification on this dataset. We adopt the MobileNet architecture from Garyfallos et al. (2019) as the foundation for our experiments. MobileNet is a Convolutional Neural Network (CNN) specifically designed for image classification and mobile vision applications. It employs depthwise separable convolutions to significantly reduce the number of parameters while maintaining sufficient depth and accuracy. These computational savings make MobileNet well-suited for deployment on low-power devices such as cellphones and embedded systems, and thus a practical choice for implementation in healthcare settings.

Direct: The direct approach attaches a softmax layer directly to the MobileNet output to produce an N-dimensional vector $q_j^d(\beta_d, X)$. Parameters β_d are optimized to minimize the empirical waiting cost over training samples, as described in §4.

Type-First: A MobileNet classifier (see Appendix §7.2) predicts a T-dimensional probability vector $t(X_s)$ for each image s with feature vector X_s, where $t_i(X_s)$ is the predicted probability that image s is of type i. Then, we assign image s to queue $j \in \{1,2,3,4\}$ with probability:

$$q_j^t(\beta_t, t(X)) = \begin{cases} \frac{\exp(\beta_{tj} \cdot t(X))}{\sum_{k=1}^{3} \exp(\beta_{tk} \cdot t(X)) + 1}, & j = 1,2,3 \\ 1 - \sum_{k=1}^{3} q_k^t(\beta_t, t(X)), & j = 4 \end{cases}$$

where each $\beta_{tj} \in \mathbb{R}^T$ is optimized to minimize the estimated waiting cost, as described in §4.

Performance Benchmarks: To benchmark the performance of direct and type-first, we estimate the average waiting cost under perfect type information with classification matrix $P^{perfect} = I^*_{T \times N}$ (see eq. (5) in §2). We also consider the single queue first-in-first-out benchmark with classification matrix P^{FIFO}: $P_{ij}^{FIFO} = 1 \; \forall i \in \mathcal{T}, j = 1$

5.2. Results

Average Waiting Costs: Table 3 reports the mean and standard deviation of average waiting costs across 10 samples of 2000 test images under the different approaches described above. Across all delay-cost structures, direct outperforms type-first, often halving waiting costs. Although type-first improves upon FIFO, it performs much worse than the perfect type information benchmark—especially in the steep cost settings (see rows 2 and 4), where direct closes much of the optimality gap.

Average Waiting Time by Type: To understand the sources of waiting cost differences, Table 4 in the Appendix reports the average waiting time for each type under different queue assignment approaches and delay-cost structures. We observe the following patterns:

- The direct approach allocates delay more effectively—ensuring that high-cost types receive prompt service while low-cost types tolerate longer waits—closely mirroring the perfect type information benchmark. In contrast, the type-first approach achieves weaker separation between high- and

Table 3 **Mean (standard deviation) of average waiting costs across 10 samples of 2000 test images.**

Delay-Cost Structure	$\mathcal{C}(\widehat{P}^t)$	$\mathcal{C}(\widehat{P}^d)$	$\mathcal{C}(P^{\text{perfect}})$	$\mathcal{C}(P^{\text{FIFO}})$
$c_i = 1.8(T-i)+1$	58 (8)	27 (1)	16	59
$c_i = 10(T-i)+1$	270 (46)	106 (10)	51	290
$c_i = 1.5^{(T-i)}$	103 (18)	59 (4)	28	166
$c_i = 1.8^{(T-i)}$	550 (50)	395 (41)	158	1285

low-cost types, especially when delay-cost heterogeneity is low. For example, under $c_i = 1.8(T-i)+1$, the average waiting times for types 1 and 14 are 2.69 and 12.25, respectively, under the direct approach, and 3.84 and 8.83, respectively, under the type-first approach.

- The type-first approach often misprioritizes intermediate-cost types (e.g., types 2–6), resulting in high average waiting times—sometimes even exceeding those of the lowest-cost type (type 14), particularly when delay-cost heterogeneity is low. For instance, under $c_i = 1.8(T-i)+1$, the average waiting times for types 2 to 6 range from 8.59 to 13.71, while the waiting time for type 14 is 8.83.

The Role of Delay Cost Structures: Because the number of types exceeds the number of queues, the delay cost structure impacts the optimal grouping of types into queues (see Propositions 2 and 6). Table 4 shows that under perfect type information, the average waiting time for type 13 increases from 2.13 to 13.41 minutes while that of type 1 decreases from 0.93 to 0.87 minutes as the delay cost structure goes from linear to convex and it becomes optimal to isolate the highest cost types. Delay costs can therefore serve as design parameters which can be tuned to achieve desired waiting times.

Robustness of the Direct Approach: Proposition 6 shows that optimal queue assignment under noisy features depends jointly on the cost structure (captured by γ) and the statistical distances between feature distributions. The direct approach re-optimizes its classifier parameters end-to-end against the waiting cost objective for each cost structure, allowing it to jointly capture this interplay in the feature space. In contrast, the type-first approach locks in a cost-agnostic type classifier in stage 1; only the queue mapping in stage 2 adapts to costs, but it operates on the classifier's outputs rather than the underlying features. This limits its ability to adjust to different cost structures, particularly when the type classifier deviates from the Bayes posterior. Table 4 confirms this: waiting times under type-first are more sensitive to the cost structure than under direct. For instance, the waiting time for Type 1 ranges from 2.33 to 4.69 under type-first but only from 2.10 to 2.69 under direct across cost settings.

Classification Matrices: Since average waiting times in queues depend on the underlying classification probabilities, we compute the averages of the classification matrices under different queue assignment approaches to better understand the differences in their performance. Equation (28) re-

ports the averages of the classification matrices—computed over 10 samples of 2000 test images—for type-first ($\widehat{P}^t$), direct ($\widehat{P}^d$), and perfect type information ($\widehat{P}^{\text{perfect}}$) under the cost setting $c_i = 1.8^{(T-i)}$.

$$\widehat{P}^t = \begin{bmatrix} 0.69 & 0.24 & 0.08 & 0.0 \\ 0.38 & 0.40 & 0.22 & 0.0 \\ 0.27 & 0.46 & 0.27 & 0.0 \\ 0.17 & 0.52 & 0.31 & 0.0 \\ 0.21 & 0.39 & 0.40 & 0.0 \\ 0.14 & 0.40 & 0.47 & 0.0 \\ 0.67 & 0.27 & 0.06 & 0.0 \\ 0.34 & 0.51 & 0.16 & 0.0 \\ 0.48 & 0.48 & 0.03 & 0.0 \\ 0.42 & 0.47 & 0.11 & 0.0 \\ 0.54 & 0.39 & 0.07 & 0.0 \\ 0.25 & 0.47 & 0.28 & 0.0 \\ 0.23 & 0.55 & 0.22 & 0.0 \\ 0.05 & 0.24 & 0.71 & 0.0 \end{bmatrix}, \quad \widehat{P}^d = \begin{bmatrix} 0.0 & 0.83 & 0.14 & 0.03 \\ 0.0 & 0.77 & 0.19 & 0.05 \\ 0.0 & 0.78 & 0.17 & 0.05 \\ 0.0 & 0.75 & 0.21 & 0.05 \\ 0.0 & 0.63 & 0.27 & 0.10 \\ 0.0 & 0.61 & 0.28 & 0.10 \\ 0.0 & 0.81 & 0.15 & 0.04 \\ 0.0 & 0.62 & 0.30 & 0.08 \\ 0.0 & 0.63 & 0.27 & 0.10 \\ 0.0 & 0.67 & 0.25 & 0.09 \\ 0.0 & 0.71 & 0.21 & 0.08 \\ 0.0 & 0.63 & 0.26 & 0.11 \\ 0.0 & 0.64 & 0.27 & 0.09 \\ 0.0 & 0.16 & 0.31 & 0.53 \end{bmatrix}, \quad \widehat{P}^{\text{perfect}} = \begin{bmatrix} 1.0 & 0.0 & 0.0 & 0.0 \\ 1.0 & 0.0 & 0.0 & 0.0 \\ 1.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 1.0 & 0.0 & 0.0 \\ 0.0 & 1.0 & 0.0 & 0.0 \\ 0.0 & 1.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 1.0 \\ 0.0 & 0.0 & 0.0 & 1.0 \end{bmatrix} \tag{28}$$

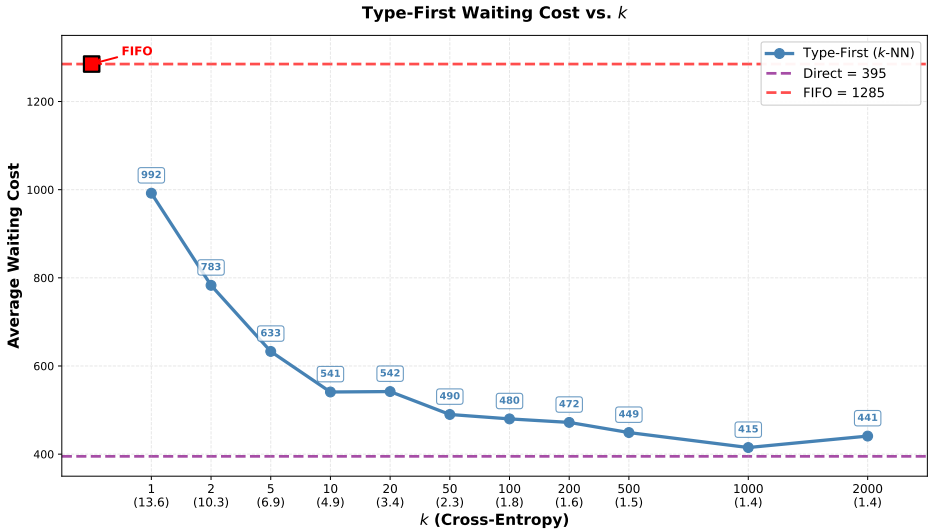
The classification matrices reveal key insights on optimal queue design, some established in §3.3.1.

1. **Statistical Pooling:** The matrices $\widehat{P}^d$ and $\widehat{P}^t$ show that both the direct and type-first approaches utilize only three out of the four queues. In Proposition 6, we establish that this phenomenon can arise under high overlap of feature distributions.
2. **Over-triage of the Medium Type:** The direct approach has a significant over-triage of the medium types: 61% of the medium type 6 join the first queue and 10% join the last queue. In contrast, under the suboptimal type-first, 14% of type 6 join the first queue and 47% join the last.
3. **Sensitivity more important than Specificity:** The direct approach correctly assigns 83% of type 1 to the first queue. However, the accuracy of correctly assigning type 14 to the last queue is only 53%. In contrast, the type-first approach has approximately the same accuracy in classifying type 1 to the first queue and type 14 to the last queue (69% and 71%, respectively).

5.3. Sensitivity to Type Classifier Quality

Proposition 3 and Example 3 suggest that type-first may match the direct performance when the type classifier closely approximates the Bayes posterior. Since k-nearest neighbors (k-NN) classification converges to the Bayes posterior as the number of neighbors $k \to \infty$ and the training size $S \to \infty$ with $k/S \to 0$ (Bishop 2006), we leverage it to empirically test this hypothesis using our chest X-ray dataset. We use the 512-dimensional feature embeddings from the penultimate layer of the trained MobileNet (used for type-first in the previous experiments), which provide a robust, low-dimensional representation of the high-dimensional raw image pixels. We train k-NN classifiers on 80% of the data and evaluate on the remaining 20% test set using the cost structure $c_i = 1.8^{(T-i)}$ with $\rho = 0.9$.

Figure 3 **Sensitivity to Type Classifier Quality. Average waiting cost versus neighborhood size k. The values in parentheses on the x-axis indicate the test set cross-entropy; lower values imply a closer approximation to the Bayes posterior. Cost structure: $c_i = 1.8^{(T-i)}$, $\rho = 0.9$.**



We employ the cross-entropy between k-NN predictions and true type labels as a proxy for deviation from the Bayes posterior. By varying k from 1 to 2000, we systematically modulate this deviation.

Figure 3 shows the average waiting cost under type-first for different values of k and the corresponding cross-entropies (in parentheses). As k increases from 1 to 1000, cross-entropy decreases from 13.6 to 1.44, and cost decreases monotonically from 992 to 415. At $k = 1000$, type-first achieves a cost of 415, within 5% of direct (395). The slight cost increase at k=2000 reflects oversmoothing: as k grows large relative to the fixed dataset size, the classifier averages over nearly all observations rather than using local information, and in the limit predicts the same value for every observation.

The empirical evidence is consistent with the convergence of the type-first performance to that of the direct method as the classifier converges to the Bayes posterior.

As a practical solution, the k-NN is computationally expensive. It requires retaining the full training embeddings and performing neighbor searches at inference. In contrast, our original type-first and direct implementations use the same lightweight MobileNet architecture, with the direct approach leading to significantly better performance.

Study Limitations and Possible Extensions

The performance reported in Table 3 is likely a conservative estimate of the gains from the direct approach because our neural network design was not tuned for queue prediction. We did not optimize certain design parameters of the MobileNet neural network—such as the arrangement of dropout layers, dropout rate, and learning rate—which were originally tuned for disease prediction in Garyfallos

et al. (2019). Since we repurpose the network for priority queue prediction, tuning these parameters specifically for our objective could lead to further improvements in performance.

While this computational study provides strong evidence that the direct approach yields substantial gains over the type-first approach in the context of chest X-ray triage, several modeling simplifications limit its scope. First, our dataset does not include contextual features, and we therefore assume that queue assignments depend only on image features. In practice, both clinical (e.g., patient history) and non-clinical (e.g., referring physician) factors may influence queue assignment decisions. Second, prior studies (Ibanez et al. 2018) have shown that radiologists often override system-assigned priorities, exercising discretion in real-world triage.

Finally, we assume that delay-costs accumulate linearly over time. In clinical practice the waiting cost for critical conditions may be non-linear, potentially exhibiting sharp increases beyond specific thresholds where survival probabilities degrade rapidly. In our linear-cost model, the cost coefficients serve as design proxies for this urgency. By assigning sufficiently high delay costs to critical types, practitioners can constrain waiting times to remain below these safety thresholds, effectively mitigating the risks associated with non-linear deterioration. Nevertheless, exploring queue assignment policies that directly optimize non-linear objectives (e.g., Lee et al. 2024) remains a promising extension and may yield counter-intuitive results, as illustrated in Section 9 of Argon and Ziya (2009).

6. Summary, Discussion and Conclusion

Priority queuing is an unexplored context for integrated prediction and optimization. The nonlinear queuing externalities—the impact of priority assignment of one job on the waiting times of other jobs—give rise to new questions about the interaction between classification/prediction and priority queues (workflow) design that warrant a detailed study.

Our study highlights the importance of moving beyond the traditional type-first approach to a direct approach for minimizing waiting costs. The direct approach, which optimizes priority queue probabilities directly from features, outperforms the type-first method unless the type classifier perfectly predicts the Bayes posterior. Our analysis, both theoretical and numerical, demonstrates that the type-first approach may fail to converge to the optimal solution in scenarios involving model misspecification. In contrast, the direct approach can appropriately optimize the parameters to minimize the average waiting cost, and achieve a near optimal solution even with a misspecified classification model. Further, we also show that the direct approach is computationally feasible.

Our theoretical analysis of the three-type setting helps characterize the mechanisms through which direct queuing brings improvements under different levels of system utilization and relative urgency of the types: 1) balancing under and over prioritization of the medium cost types, 2) optimizing the classification accuracy for the most important types, and 3) accounting for statistical distances

between feature distributions while optimizing the utilizations of the three queues, with fewer queues (than the maximum available) utilized under high overlap of feature distributions. These novel results provide interpretability to the mathematical interaction between AI/ML (classification) and priority queuing.

We also present evidence that the direct approach can yield substantial improvements over the type-first approach in a practical setting of triaging chest x-rays. Our computational performance study may be the first to apply deep learning-based classification of medical images to study the optimal feature-driven priority queue design. We show that feature-driven priority queuing can improve waiting cost significantly under high utilization by focusing on the correct queue classification for the most critical delay-cost types, and by prioritizing sensitivity over specificity.

Our research proposes a priority recommendation system and demonstrates its value in a stylized setting assuming the user follows the priority recommendation. Accounting for customer choice and human deviations from algorithmic recommendations is undoubtedly an exciting problem addressed in some other contexts (Sun et al. 2022a) but beyond the scope of this first study on feature-driven design of priority queues.

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7. Appendix

7.1. Results of Numerical Experiments in §5

Table 4 **Average waiting time for each type when $\rho = 0.9$ and c_i is the delay-cost per unit time of type i**

	$c_i = 1.8(T-i)+1$				$c_i = 10(T-i)+1$			
	Type-First	Direct	Perfect	FIFO	Type-First	Direct	Perfect	FIFO
Type 1	3.84	2.69	0.93	8.1	4.69	2.25	0.93	8.1
Type 2	8.59	2.47	0.93	8.1	9.22	2.31	0.93	8.1
Type 3	10.01	2.61	0.93	8.1	10.43	2.54	0.93	8.1
Type 4	9.07	2.58	0.93	8.1	9.63	2.45	0.93	8.1
Type 5	11.39	3.23	0.93	8.1	11.83	2.99	0.93	8.1
Type 6	13.71	3.84	1.43	8.1	14.94	3.37	1.43	8.1
Type 7	3.73	2.85	1.43	8.1	4.46	2.31	1.43	8.1
Type 8	4.89	3.9	1.43	8.1	5.00	3.75	1.43	8.1
Type 9	1.86	3.65	2.13	8.1	1.85	3.62	2.13	8.1
Type 10	3.37	3.67	2.13	8.1	3.61	3.65	2.13	8.1
Type 11	2.63	3.47	2.13	8.1	2.76	3.45	2.13	8.1
Type 12	7.95	3.59	2.13	8.1	7.91	3.57	2.13	8.1
Type 13	5.82	3.42	2.13	8.1	6.79	3.42	2.13	8.1
Type 14	8.83	12.25	13.82	8.1	8.25	12.54	13.82	8.1
	$c_i = 1.5^{(T-i)}$				$c_i = 1.8^{(T-i)}$			
	Type-First	Direct	Perfect	FIFO	Type-First	Direct	Perfect	FIFO
Type 1	2.40	2.14	0.87	8.1	2.33	2.10	0.87	8.1
Type 2	6.08	2.62	0.87	8.1	4.43	2.64	0.87	8.1
Type 3	6.37	2.16	0.87	8.1	5.20	2.62	0.87	8.1
Type 4	7.33	3.12	1.10	8.1	5.79	2.63	1.10	8.1
Type 5	9.30	3.09	1.10	8.1	6.94	3.81	1.10	8.1
Type 6	10.71	3.76	1.10	8.1	7.91	3.94	1.10	8.1
Type 7	2.12	2.28	1.69	8.1	2.06	2.33	1.69	8.1
Type 8	3.99	4.67	1.69	8.1	3.66	3.58	1.69	8.1
Type 9	1.71	5.67	1.69	8.1	1.89	3.85	1.69	8.1
Type 10	2.63	4.49	1.69	8.1	2.94	3.49	1.69	8.1
Type 11	2.15	3.80	1.69	8.1	2.31	3.26	1.69	8.1
Type 12	5.97	4.30	1.69	8.1	5.33	3.93	1.69	8.1
Type 13	4.61	4.22	13.41	8.1	4.64	3.64	13.41	8.1
Type 14	10.41	12.27	13.41	8.1	11.25	12.36	13.41	8.1

7.2. Type-Classification for Triage of Chest X-rays

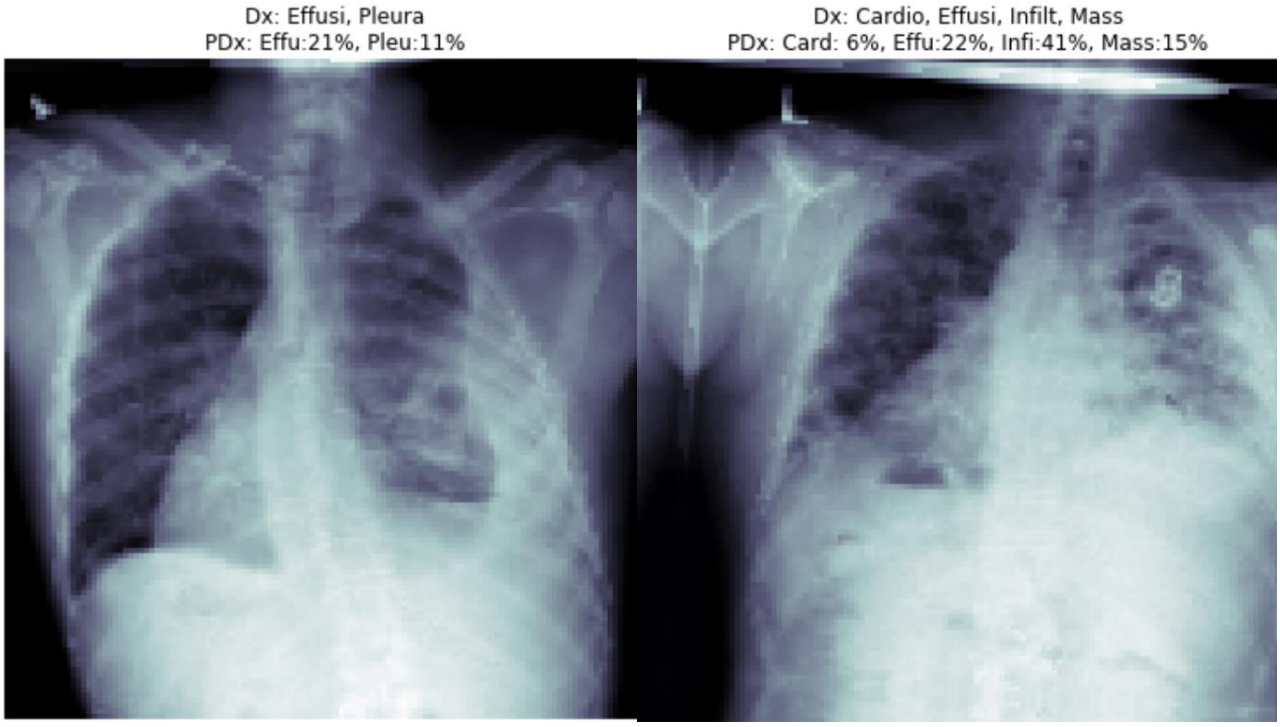

Figure 4 **2D chest X-rays labeled with diseases and Mobile Net predictions. Dx: disease findings extracted from their radiologist reports. PDx: predicted probabilities by Mobile Net*. Image Source: Garyfallos et al. (2019).**

*Mobile Net outputs probabilities for all 14 findings but the figure shows only those for the actual findings.

We use a MobileNet architecture (with multiple binarizers as in Garyfallos et al. (2019))[1]—a lightweight convolutional neural network (CNN)—to predict the probability of each disease. Figure 4 shows two chest X-rays (CXRs) labeled with disease findings extracted from radiologist reports, along with the corresponding MobileNet predictions. The model is trained to minimize binary cross-entropy loss.

Figure 5 reports the performance of MobileNet. For all diseases, the results—measured by the area under the receiver operating characteristic curve (AUC-ROC)—closely match those reported in Garyfallos et al. (2019) and in the earlier study by Wang et al. (2017) on chest X-ray classification.

The probability of type i corresponds to the probability that the patient has the disease with rank order i but does not have any disease with a higher rank. For example, because 1 is the highest rank, the predicted probability that an image is of type 1 is simply the predicted probability that the image has Pneumothorax:

$$\mathbb{P}(\text{type} = 1) = \mathbb{P}(\text{Pneumothorax}).$$

[1] For training the MobileNet, we create a re-sampled dataset that is well-balanced across the types.

The predicted probability that an image is of type 2 is the probability that it has Emphysema *and not* Pneumothorax:

$$\mathbb{P}(\text{type} = 2) = \mathbb{P}(\text{Emphysema}) \cdot (1 - \mathbb{P}(\text{Pneumothorax})) \,.$$

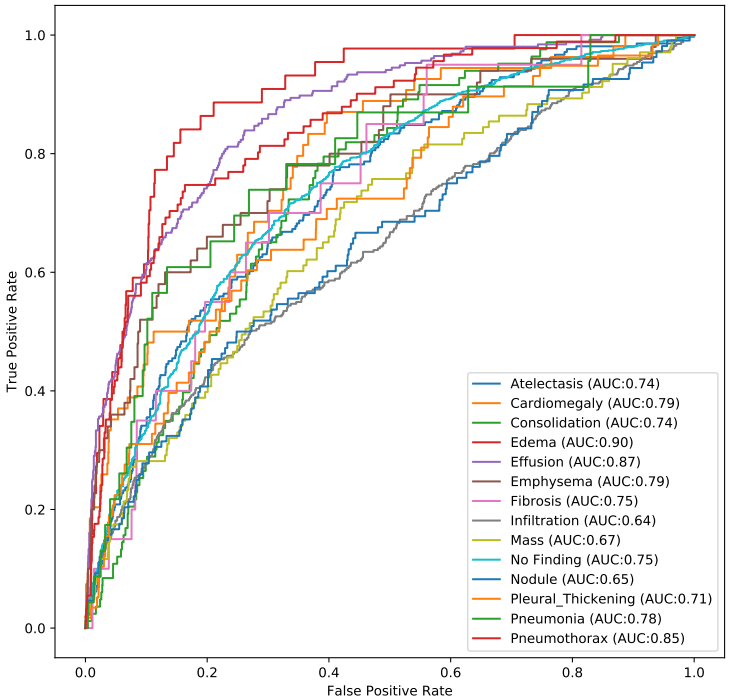

Figure 5 ROC curves for disease classification for a sample of 2000 images from the ChestX-ray14 dataset.

7.3. Example: Dominance of Direct with the same Classifier as in Type-First

The example below shows that using the same classifier to predict queue-assignment probabilities in the direct implementation can yield a lower average waiting cost than using it to predict type probabilities in type-first.

Example 4 *When $T = N = 2$, the average waiting cost $\mathcal{C}(P)$ can be written as a function of P_{11} and P_{21} (see equation (36) in Appendix 7.8):*

$$\mathcal{C}(P) = \frac{(c_1\mu_1\rho_1 + c_2\mu_2\rho_2)\,\mathbb{E}[\mathcal{R}]}{1-\rho} - \left(\frac{(c_1\mu_1 - c_2\mu_2)\,\mathbb{E}[\mathcal{R}]\,\rho_1\rho_2}{1-\rho}\right)\cdot\frac{1-\Psi(P)}{\rho_2+\rho_1\Psi(P)},$$

where

$$\Psi(P) := \frac{1-\rho P_{11}}{1-\rho P_{21}}. \tag{29}$$

Since $\mathcal{C}(P)$ is increasing in $\Psi(P)$, minimizing $\mathcal{C}(P)$ is equivalent to minimizing $\Psi(P)$. Let Ω denote the optimal value of $\Psi(P)$ over all measurable mappings $\sigma : \mathcal{X} \to [0,1]$ interpreted as queue-1 probabilities:

$$\Omega = \min_{\sigma}\ \frac{1-\rho\,\mathbb{E}_{f_1}[\sigma(X)]}{1-\rho\,\mathbb{E}_{f_2}[\sigma(X)]}, \qquad \mathbb{E}_{f_i}[\sigma(X)] = \int_{\mathcal{X}} \sigma(X) f_i(X)\,dX.$$

Fix a classification architecture $\sigma(\theta,\cdot)$.

Direct. *Using $\sigma(\theta,\cdot)$ as the direct queue classifier $q_1^d(\beta_d, X) = \sigma(\theta, X)$ yields*

$$\Psi\big(P^d(\theta)\big) = \frac{1-\rho\,\mathbb{E}_{f_1}[\sigma(\theta,X)]}{1-\rho\,\mathbb{E}_{f_2}[\sigma(\theta,X)]}.$$

Thus direct minimizes $\Psi(P^d(\theta))$ by choosing θ. If the class $\{\sigma(\theta,\cdot)\}$ can vary $\big(\mathbb{E}_{f_1}[\sigma(\theta,X)], \mathbb{E}_{f_2}[\sigma(\theta,X)]\big)$ continuously on the set of all possible θ such that there exists θ for which

$$1-\rho\,\mathbb{E}_{f_1}[\sigma(\theta,X)] = \big(1-\rho\,\mathbb{E}_{f_2}[\sigma(\theta,X)]\big)\,\Omega,$$

then $\Psi(P^d(\theta))$ achieves the minimum of $\Psi(P)$, and therefore the direct approach achieves the minimum possible average waiting cost over all feature-to-queue mappings.

Type-first. *Using the same architecture as a type classifier $z = \sigma(\theta_t, X)$ with a queue map $q^t(\beta_t,\cdot)$, and using the type-classifier-based classification probabilities, type-classifier-based ratio equals*

$$\Psi\Big(\widetilde{P}^t(\beta_t)\Big) = \frac{1-\rho\cdot\dfrac{\mathbb{E}_f\big[\sigma(\theta_t,X)\,q_1^t(\beta_t,\sigma(\theta_t,X))\big]}{\mathbb{E}_f[\sigma(\theta_t,X)]}}{1-\rho\cdot\dfrac{\mathbb{E}_f\big[(1-\sigma(\theta_t,X))\,q_1^t(\beta_t,\sigma(\theta_t,X))\big]}{\mathbb{E}_f[(1-\sigma(\theta_t,X))]}}, \tag{30}$$

whereas the theoretical ratio equals

$$\Psi(P^t(\beta_t)) = \frac{1-\rho\,\mathbb{E}_{f_1}\big[q_1^t(\beta_t,\sigma(\theta_t,X))\big]}{1-\rho\,\mathbb{E}_{f_2}\big[q_1^t(\beta_t,\sigma(\theta_t,X))\big]} = \frac{1-\rho\cdot\dfrac{\mathbb{E}_f\big[\mathbb{P}(1\mid X)\,q_1^t(\beta_t,\sigma(\theta_t,X))\big]}{\mathbb{E}_f[\mathbb{P}(1\mid X)]}}{1-\rho\cdot\dfrac{\mathbb{E}_f\big[\mathbb{P}(2\mid X)\,q_1^t(\beta_t,\sigma(\theta_t,X))\big]}{\mathbb{E}_f[\mathbb{P}(2\mid X)]}}. \tag{31}$$

Hence type-first recovers the direct optimum only if: (i) the minimizer of $\Psi(\widetilde{P}^t(\beta_t))$ *also minimizes* $\Psi(P^t(\beta_t))$*, and (ii) there exists* β_t *with*

$$1-\rho\,\mathbb{E}_{f_1}\left[q_1^t(\beta_t,\sigma(\theta_t,X))\right]=\left(1-\rho\,\mathbb{E}_{f_2}\left[q_1^t(\beta_t,\sigma(\theta_t,X))\right]\right)\Omega.$$

When $\sigma(\theta_t,X)\neq\mathbb{P}(1\mid X)$*, the objectives in* (30) *and* (31) *generally differ, so (i) can fail. Moreover, if* $\sigma(\theta_t,X)$ *takes the same value on feature subsets where* $\mathbb{P}(1\mid X)$ *differs, then every type-first policy assigns those features the same queue probabilities, whereas direct can assign them differently; this expands the set of attainable classification matrices and can strictly lower* $\Psi(P)$ *and* $\mathcal{C}(P)$*.*

7.4. Linear Formulation for Optimizing a Direct Queue Classifier ($N=2$)

From equations (2) and (3), for the average waiting cost for a classification matrix $\widehat{P}^d(\beta_d)$ can be simplified as

$$\mathcal{C}(\widehat{P}^d(\beta_d))=\mathbb{E}[\mathcal{R}]\sum_{j=1}^{N}\frac{\sum_{i=1}^{T}c_i\mu_i\rho_i\widehat{P}^d_{ij}(\beta_d)}{(1-\sum_{l<j}\rho(l,P))(1-\sum_{l\leq j}\rho(l,P))}\tag{32}$$

For $N=2$

$$\mathcal{C}(\widehat{P}^d(\beta_d))=\mathbb{E}[\mathcal{R}]\left[\frac{\sum_{i=1}^{T}c_i\mu_i\rho_i\widehat{P}^d_{i1}(\beta_d)}{(1-\sum_{i=1}^{T}\rho_i\widehat{P}^d_{i1}(\beta_d))}+\frac{\sum_{i=1}^{T}c_i\mu_i\rho_i(1-\widehat{P}^d_{i1}(\beta_d))}{(1-\sum_{i=1}^{T}\rho_i\widehat{P}^d_{i1}(\beta_d))(1-\rho)}\right]\tag{33}$$

$$\mathcal{C}(\widehat{P}^d(\beta_d))=\mathbb{E}[\mathcal{R}]\left[\frac{\sum_{i=1}^{T}c_i\mu_i\rho_i-\rho\sum_{i=1}^{T}c_i\mu_i\rho_i\widehat{P}^d_{i1}(\beta_d)}{(1-\sum_{i=1}^{T}\rho_i\widehat{P}^d_{i1}(\beta_d))(1-\rho)}\right]=\frac{\mathbb{E}[\mathcal{R}]K}{1-\rho}\left[\frac{1-\frac{\rho}{K}\sum_{i=1}^{T}c_i\mu_i\rho_i\widehat{P}^d_{i1}(\beta_d)}{(1-\sum_{i=1}^{T}\rho_i\widehat{P}^d_{i1}(\beta_d))}\right]\tag{34}$$

where

$$\widehat{P}^d_{i1}(\beta_d)=\frac{\sum_{s=1}^{S}y_{is}q_1^d(\beta_d,X_S)}{S_i},\quad K=\sum_{i=1}^{T}c_i\mu_i\rho_i,\quad S_i=\sum_{s=1}^{S}y_{is},\quad \mathbb{E}[\mathcal{R}]=\sum_{i=1}^{T}\frac{\lambda_i}{\mu_i^2}.$$

When $q_1^d(\beta_d,X)$ is linear in X, $\mathcal{C}(\widehat{P}^d(\beta_d))$ can be minimized via linear programming.

Lemma 3 (The Linear Classifier) *Suppose* $q_1^d(\beta_d,X)=\beta_d\cdot X$*, and* $\overline{X}_i$ *denotes the average value of the feature vector* X *for the jobs of type* i*:*

$$\overline{X}_i=\frac{\sum_{s=1}^{S}y_{is}X_{is}}{\sum_{s=1}^{S}y_{is}}.$$

Then, the optimal parameter vector $\beta_d=\frac{1}{w_0}w$ *and* (w_0,w) *solve the following linear program:*

$$\begin{aligned}\min_{w_0\in\mathcal{R},w\in\mathcal{R}^p}\quad & w_0-\frac{\rho}{K}w\cdot\sum_{i=1}^{T}c_i\mu_i\rho_i\overline{X}_i\\ s.t.\quad & w_0-w\cdot\sum_{i=1}^{T}\rho_i\overline{X}_i=1\\ & 0\leq w\cdot X_s\leq w_0\quad\forall s\in\mathcal{S}\end{aligned}$$

Proof of Lemma 3. The linear program can be obtained by substituting $q_1^d(\beta_d, X_s) = \beta_d \cdot X_s$ in (34):

$$\widehat{P}_{i1}^d(\beta_d) = \frac{\sum_{s=1}^S y_{is} q_1^d(\beta_d, X_s)}{S_i} = \beta_d \cdot \bar{X}_i \tag{35}$$

Next, substitute

$$w_0 = \frac{1}{(1 - \sum_{i=1}^T \rho_i \widehat{P}_{i1}^d(\beta_d))}, \; w = w_0 \beta_d.$$

The constraints correspond to the classifier output $q_1^d(\beta_d, X_s)$ lying in [0,1] for each X_s.

Sensitivity Analysis. A full characterization of how the optimal direct solution varies with delay-costs, service times, or arrival rates is nontrivial. However, standard results from linear programming sensitivity analysis (Murty 1983) imply that the solution remains stable within a neighborhood of these parameters. Thus, the optimal classifier, along with the resulting classification probabilities and expected waiting times, is robust to small estimation errors.

7.5. Optimizing the Direct Queue Classifier: Practical Challenges

While Lemma 2 shows that gradient descent converges to a stationary point, two practical challenges arise when implementing it:

- For a fixed number of iterations M, convergence slows when server utilization ρ or aggregate delay cost $\sum_{i=1}^T c_i \mu_i \rho_i$ is high.
- Large datasets and high-dimensional parameters make full-batch gradient descent computationally expensive. In practice, Stochastic Gradient Descent (SGD) is used, where each update relies on a random minibatch. Since the waiting cost depends on empirical estimates of $T \times N$ classification probabilities, minibatches must be sufficiently representative. A larger $T \times N$ thus requires larger batch sizes, a challenge shared with type-first training for large T.

Advances in numerical processing and computing power suggest that these challenges will not pose fundamental barriers to implementing direct queue classifiers.

7.6. Type Classifier Implementations

Parametric Classifiers: These include Maximum Likelihood Estimation (MLE), Bayesian Estimation, and the Method of Moments (MoM). These are standard classifiers and we spell these out for clarity on their use in type classification.

1. **MLE** assumes a parametric functional form for $t(X)$ (e.g., softmax or neural network) and fits parameters by maximizing the likelihood (Millar 2011):

$$t(X) = t(\theta, X); \quad \theta^* = \arg\max_\theta \sum_{s=1}^S \sum_{i=1}^T y_{is} \log(t_i(\theta, X_s)).$$

2. **Bayesian Estimation** assumes parametric models for the type-conditional feature densities $f_i(X) = f_i(\theta_i, X)$, and places a joint prior distribution $\pi(\theta)$ over the full parameter vector $\theta = (\theta_1, \ldots, \theta_T)$. Given the dataset $\mathcal{H}_S$, the posterior over θ is given by:

$$\pi(\theta \mid \mathcal{H}_S) \propto \left[\prod_{s=1}^{S} \prod_{i=1}^{T} f_i(\theta_i, X_s)^{y_{is}} \right] \pi(\theta).$$

The predicted type posterior $t_i(X)$ for a feature X is computed as:

$$t_i(X) = \frac{p_i \int f_i(\theta_i, X)\, \pi(\theta \mid \mathcal{H}_S)\, d\theta}{\sum_{k=1}^{T} p_k \int f_k(\theta_k, X)\, \pi(\theta \mid \mathcal{H}_S)\, d\theta}.$$

Here, p_i denotes the prior probability of type i, and the integrals are typically evaluated using Monte Carlo methods (Gelman et al. 2013, Bernardo and Smith 2009).

3. **Method of Moments (MoM)** also assumes parametric models for the type-conditional densities $f_i(X) = f_i(\theta_i, X)$. The parameter estimates $\hat{\theta}_i$ are obtained by solving moment-matching conditions:

$$t_i(X) = \frac{p_i f_i(\theta_i, X)}{\sum_{k=1}^{T} p_k f_k(\theta_k, X)}; \quad \frac{\sum_{s=1}^{S} y_{is} X_s^{\otimes r}}{\sum_{s=1}^{S} y_{is}} = \mathbb{E}_{X \sim f_i(\theta_i, \cdot)}[X^{\otimes r}], \quad \forall i \in \mathcal{T},\ r = 1, \ldots, R_i,$$

where $X_s^{\otimes r}$ denotes the rth-order tensor (outer) product of the feature vector $X_s = (x_{s1}, x_{s2}, \ldots, x_{sp})$. An arbitrary entry of this tensor, indexed by $(j_1, j_2, \ldots, j_r)$, equals

$$[X_s^{\otimes r}]_{j_1, j_2, \ldots, j_r} = x_{s,j_1} \cdot x_{s,j_2} \cdots x_{s,j_r}, \quad \text{for all } j_k \in \{1, \ldots, p\}.$$

The number of moment conditions R_i is typically chosen to match the number of free parameters in θ_i, ensuring identifiability (Greene 2018).

Non-Parametric Classifiers: k-Nearest Neighbors (k-NN) estimates the type probabilities without assuming a functional form. For all $i \in \mathcal{T}$, the predicted type-probability:

$$t_i(X) = \frac{1}{k} \sum_{s \in \mathcal{S}_k(X)} y_{is},$$

where $\mathcal{S}_k(X)$ denotes the indices of the k nearest neighbors of X.

Limitations of common type classifiers in Bayes posterior convergence. Maximum likelihood estimators converge only under correct parametric specification and regularity (e.g., identifiability); performance degrades under misspecification, high dimension, limited data, or nonconvex landscapes (van der Vaart 1998). Bayesian estimators converge when the generative models for the type-conditional feature densities are correctly specified, but in practice accurate model specification (and priors) is difficult in complex, high-dimensional domains (Gelman et al. 2013). Method-of-moments estimators require an injective parameter-to-moment map and moment-determinate distributions; multimodality, heavy tails, or weak/incomplete moments can lead to inconsistency, with

risks exacerbated in high dimensions (Greene 2018). Finally, k-nearest neighbors is consistent if the Bayes posterior is sufficiently smooth and $k \to \infty$ with $k/S \to 0$ as sample size $S \to \infty$, but convergence is slow and deteriorates with dimension due to the curse of dimensionality (Härdle et al. 2004, Hastie et al. 2009).

7.7. Numerical Example 3

Example 3 (Direct vs. Type-First for a One-Dimensional Feature) *We compare the direct and type-first approaches in a stylized setting with two types ($T=2$) and two queues ($N=2$). Type-1 is rare but costly ($p_1=0.1$, $c_1>c_2$); both types have equal service rates ($\mu_1=\mu_2$), and the system's total utilization is $\rho=0.7$. The type of a job is unobserved at arrival and must be inferred from a one-dimensional feature X, which follows a uniform distribution on $[0,\frac{3}{2}]$ for type 1 and $[-1,1]$ for type 2. The optimal policy under known feature distributions assigns all jobs with $X \geq 0$ to queue 1, yielding optimal classification probabilities: $P_{11}^{opt}=1$, $P_{12}^{opt}=0$, $P_{21}^{opt}=0.5$, and $P_{22}^{opt}=0.5$.*

First, we will evaluate the optimal feature-driven prioritization policy for this example.

The Bayes posterior probability over the domain of X, i.e., $[-1,\frac{3}{2}]$, is:

$$\mathbb{P}(1 \mid X) = \begin{cases} 0 & \text{if } -1 \leq X < 0, \\ \frac{4}{31} & \text{if } 0 \leq X \leq 1, \\ 1 & \text{if } 1 < X \leq \frac{3}{2}. \end{cases}$$

The optimal policy assigns jobs with higher values of $\mathbb{P}(1 \mid X)$ to the high-priority queue (see Argon and Ziya (2009)). Since $\mathbb{P}(1 \mid X)$ is increasing in X, the optimal solution is to assign all $X \geq K$ to queue 1. (This can also be established using an interchange argument.) This results in:

$$P_{11} = 1 - F_1(K), \quad P_{21} = 1 - F_2(K).$$

As shown in equation (36) in §7.8 of this Appendix, the optimal threshold K solves the following optimization problem:

$$\min_K \frac{1-\rho(1-F_1(K))}{1-\rho(1-F_2(K))}.$$

Substituting $\rho=0.7$ and the distribution functions $F_1(K)$ and $F_2(K)$:

$$F_1(K) = \begin{cases} 0 & K<0 \\ \frac{2K}{3} & K \in [0,\frac{3}{2}] \\ 1 & K > \frac{3}{2} \end{cases}, \; F_2(K) = \begin{cases} 0 & K<-1 \\ \frac{K+1}{2} & K \in [-1,1] \\ 1 & K>1 \end{cases},$$

we obtain that the optimal value is $K=0$, yielding $P_{11}=1$ and $P_{21}=0.5$.

Next, we illustrate how we implement the different type classifiers whose performance is summarized in Table 1.

Bayesian Estimation (Assuming Gaussian Feature Distributions). We adopt the Bayesian estimation framework described in §7.6, where the type-conditional feature distribution is modeled as Gaussian:

$$X \mid \text{type} = i \sim \mathcal{N}(\mu_i, \sigma_i^2).$$

The corresponding density is

$$f_i((\mu_i, \sigma_i^2), X) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(X-\mu_i)^2}{2\sigma_i^2}\right).$$

Prior. We place a conjugate normal–inverse-gamma prior on the parameters $\theta_i = (\mu_i, \sigma_i^2)$:

$$\sigma_i^2 \sim \text{Inv-Gamma}(\alpha_0, \beta_0), \qquad \mu_i \mid \sigma_i^2 \sim \mathcal{N}(\mu_0,\, \sigma_i^2/\kappa_0).$$

Here, α_0, β_0 are prior shape and scale parameters for the variance, μ_0 is the prior mean, and κ_0 can be interpreted as a prior sample size that reflects the strength of belief in μ_0. The inverse-gamma density is

$$f(\sigma^2) = \frac{\beta_0^{\alpha_0}}{\Gamma(\alpha_0)} (\sigma^2)^{-\alpha_0-1} \exp\left(-\tfrac{\beta_0}{\sigma^2}\right), \quad \sigma^2 > 0.$$

Posterior Update. Given S_i feature observations of type i, $\mathcal{H}_S^i = \{X_s : y_{is} = 1\}$, with sample mean

$$\overline{X}_i = \tfrac{1}{S_i} \sum_{s:y_{is}=1} X_s,$$

the posterior distribution remains normal–inverse-gamma:

$$\sigma_i^2 \mid \mathcal{H}_S^i \sim \text{Inv-Gamma}(\alpha_i, \beta_i), \qquad \mu_i \mid \sigma_i^2, \mathcal{H}_S^i \sim \mathcal{N}(\mu_i^*,\, \sigma_i^2/\kappa_i),$$

with updated hyperparameters

$$\kappa_i = \kappa_0 + S_i, \quad \mu_i^* = \frac{\kappa_0\mu_0 + S_i\overline{X}_i}{\kappa_0 + S_i},$$

$$\alpha_i = \alpha_0 + \tfrac{S_i}{2}, \quad \beta_i = \beta_0 + \tfrac{1}{2} \sum_{s:y_{is}=1} (X_s - \overline{X}_i)^2 + \tfrac{\kappa_0 S_i}{2(\kappa_0+S_i)} (\overline{X}_i - \mu_0)^2.$$

Intuitively, κ_i is the effective sample size for the mean, μ_i^* is the weighted average of the prior mean and sample mean, α_i updates the degrees of freedom for the variance, and β_i updates the scale by combining within-sample variance with prior information.

Posterior Predictive. The predictive density for a new feature X is given by integrating the Gaussian likelihood with respect to the posterior over (μ_i, σ_i^2):

$$\widehat{f}_i(X) = \int f_i((\mu_i, \sigma_i^2), X)\, \pi(\mu_i, \sigma_i^2 \mid \mathcal{H}_S^i)\, d\mu_i\, d\sigma_i^2.$$

We approximate the integral via Monte Carlo. Specifically, we draw M posterior samples

$$\sigma_i^{2(m)} \sim \text{Inv-Gamma}(\alpha_i, \beta_i), \qquad \mu_i^{(m)} \sim \mathcal{N}\left(\mu_i^*, \tfrac{\sigma_i^{2(m)}}{\kappa_i}\right),$$

and approximate

$$\widehat{f}_i(X) \approx \tfrac{1}{M} \sum_{m=1}^{M} \phi(X \mid \mu_i^{(m)}, \sigma_i^{2(m)}).$$

Bayes' Rule. Finally, the estimated type posterior is

$$t_i(X) = \frac{p_i \, \widehat{f}_i(X)}{\sum_{k=1}^{T} p_k \, \widehat{f}_k(X)}.$$

Method of Moments Estimation (Assuming Uniform Feature Distributions): We adopt the method of moments framework where the type-conditional density, $f_i(\theta_i, X)$, is modeled as a uniform density with parameters $\theta_i = (a_i, b_i)$:

$$f_i((a_i, b_i), X) = \begin{cases} \frac{1}{b_i - a_i} & \text{if } X \in [a_i, b_i], \\ 0 & \text{otherwise.} \end{cases}$$

Moment Conditions. The parameters $\theta_i = (a_i, b_i)$ are estimated using the method of moments by matching the first and second empirical moments of type i samples to the theoretical moments of the uniform distribution:

$$\hat{\mu}_i = \frac{1}{S_i} \sum_{s: y_{is}=1} X_s, \quad \hat{\sigma}_i^2 = \frac{1}{S_i} \sum_{s: y_{is}=1} (X_s - \hat{\mu}_i)^2.$$

Theoretical moments for $X \sim \text{Unif}(a_i, b_i)$ are:

$$\mathbb{E}[X] = \frac{a_i + b_i}{2}, \quad \text{Var}(X) = \frac{(b_i - a_i)^2}{12}.$$

Equating these with the empirical moments yields the following closed-form estimators:

$$\hat{a}_i = \hat{\mu}_i - \sqrt{3\hat{\sigma}_i^2}, \quad \hat{b}_i = \hat{\mu}_i + \sqrt{3\hat{\sigma}_i^2},$$

where $S_i = \sum_{s=1}^{S} y_{is}$ is the number of training samples of type i.

Type-Probability Estimation. Using the estimated parameters $\hat{\theta}_i = (\hat{a}_i, \hat{b}_i)$, we compute the predicted type-probability using the Bayes rule:

$$t_i(X) = \frac{p_i \cdot f_i(\hat{\theta}_i, X)}{\sum_{k=1}^{T} p_k \cdot f_i(\hat{\theta}_i, X)}.$$

7.8. Proofs

Proof of Proposition 1: We use an interchange argument. The difference in average waiting cost per unit time on interchanging priority queues m and n is equal to (see equations 2 and 3)

$$\frac{\sum_{i\in\mathcal{T}} P_{im}\lambda_i c_i \mathbb{E}[\mathcal{R}]}{(1-\sum_{q\in[m-1]}\rho_q(P))(1-\sum_{q\in[m-1]}\rho_q(P)-\rho(m,P))} + \frac{\sum_{i\in\mathcal{T}} P_{in}\lambda_i c_i \mathbb{E}[\mathcal{R}]}{(1-\sum_{q\in[n-1]}\rho_q(P))(1-\sum_{q\in[n-1]}\rho_q(P)-\rho(n,P))}$$
$$- \frac{\sum_{i\in\mathcal{T}} P_{in}\lambda_i c_i \mathbb{E}[\mathcal{R}]}{(1-\sum_{q\in[m-1]}\rho_q(P))(1-\sum_{q\in[m-1]}\rho_q(P)-\rho(n,P))} - \frac{\sum_{i\in\mathcal{T}} P_{im}\lambda_i c_i \mathbb{E}[\mathcal{R}]}{(1-\sum_{q\in[n-1]}\rho_q(P))(1-\sum_{q\in[n-1]}\rho_q(P)-\rho(n,P))}.$$

This difference is negative iff

$$c_m(P)\mu_m(P) = \frac{\sum_{i\in\mathcal{T}} P_{im}\lambda_i c_i}{\sum_{i\in\mathcal{T}} P_{im}\lambda_i \frac{1}{\mu_i}} > \frac{\sum_{i\in\mathcal{T}} P_{in}\lambda_i c_i}{\sum_{i\in\mathcal{T}} P_{in}\lambda_i \frac{1}{\mu_i}} = c_n(P)\mu_n(P).$$

Q.E.D.

Proof of Proposition 2: We use a standard interchange argument. Toward a contradiction, suppose that the optimal partition $I^*_{T\times N}$ assigns type u to queue q and type v to queue r, where $q \le r$ (i.e., queue q has higher or equal priority than queue r), but $c_u\mu_u < c_v\mu_v$. That is, a lower-cost type is assigned to a higher-priority queue, while a higher-cost type is assigned to a lower-priority queue. Because queue indices are discrete, there must exist a pair of adjacent queues j and $j+1$ such that $q \le j < r$, and types i and k (with one of them being u or v) such that:

$$I^*_{T\times N}(i,j) = 1, \quad I^*_{T\times N}(k,j+1) = 1, \quad \text{and} \quad c_i\mu_i < c_k\mu_k.$$

Now we will generate a contradiction to the assumption that $I^*_{T\times N}$ is the optimal policy by proving that the expected stationary waiting cost is strictly lower under at least one of the two modifications to $I^*_{T\times N}$: a) shifting type i to queue $j+1$ i.e., $I^1_{T\times N}(i,j+1) = 1$; b) shifting type k to queue j i.e., $I^1_{T\times N}(k,j) = 1$. This would then contradict the optimality of $I^*_{T\times N}$.

Define

$$A := \{l : I^*_{T\times N}(l,j) = 1, l \ne i\},\ c_A\mu_A := \frac{\sum_{l\in A}\rho_l c_l \mu_l}{\sum_{l\in A}\rho_l},\ \rho_A := \sum_{l\in A}\rho_l,$$

and similarly

$$B := \{l : I^*_{T\times N}(l,j+1) = 1, l \ne k\},\ c_B\mu_B := \frac{\sum_{l\in B}\rho_l c_l \mu_l}{\sum_{l\in B}\rho_l},\ \rho_B := \sum_{l\in B}\rho_l,$$

where $c_A\mu_A = \rho_A = 0$ and $c_B\mu_B = \rho_B = 0$ if $A = \phi$ or $B = \phi$, respectively. and, finally,

$$\bar{\rho} := \sum_{l:\exists q' \le j-1 \text{ and } I^*_{T\times N}(l,q')=1} \rho_l,$$

with $\bar{\rho} = 0$ if $j \le 1$.

Then,

$$\begin{aligned}
&\mathcal{C}(I^*_{T\times N}) - \mathcal{C}(I^1_{T\times N}) \\
&\quad = \mathbb{E}[\mathcal{R}]\Bigg[\frac{c_A\mu_A\rho_A + c_i\mu_i\rho_i}{(1-\bar{\rho})(1-\bar{\rho}-\rho_A-\rho_i)} + \frac{c_k\mu_k\rho_k + c_B\mu_B\rho_B}{(1-\bar{\rho}-\rho_A-\rho_i)(1-\bar{\rho}-\rho_A-\rho_i-\rho_k-\rho_B)} \\
&\qquad\qquad - \frac{c_A\mu_A\rho_A}{(1-\bar{\rho})(1-\bar{\rho}-\rho_A)} - \frac{c_i\mu_i\rho_i + c_k\mu_k\rho_k + c_B\mu_B\rho_B}{(1-\bar{\rho}-\rho_A)(1-\bar{\rho}-\rho_A-\rho_i-\rho_k-\rho_B)}\Bigg] \\
&\quad = \rho_i\mathbb{E}[\mathcal{R}]\Bigg[\frac{c_A\mu_A\rho_A + c_i\mu_i(1-\bar{\rho}-\rho_A)}{(1-\bar{\rho})(1-\bar{\rho}-\rho_A-\rho_i)(1-\bar{\rho}-\rho_A)} \\
&\qquad\qquad + \frac{c_k\mu_k\rho_k + c_B\mu_B\rho_B - c_i\mu_i(1-\bar{\rho}-\rho_A-\rho_i)}{(1-\bar{\rho}-\rho_A-\rho_i)(1-\bar{\rho}-\rho_A)(1-\bar{\rho}-\rho_A-\rho_i-\rho_k-\rho_B)}\Bigg] \\
&\quad = \rho_i\mathbb{E}[\mathcal{R}] \cdot \frac{\Delta_1}{(1-\bar{\rho})(1-\bar{\rho}-\rho_A-\rho_i)(1-\bar{\rho}-\rho_A)(1-\bar{\rho}-\rho_A-\rho_i-\rho_k-\rho_B)},
\end{aligned}$$

where

$$\Delta_1 := c_A\mu_A\rho_A(1-\bar{\rho}-\rho_A-\rho_i-\rho_k-\rho_B)+(c_k\mu_k\rho_k+c_B\mu_B\rho_B)(1-\bar{\rho})$$
$$-c_i\mu_i(\rho_A(1-\bar{\rho}-\rho_A-\rho_i-\rho_k-\rho_B)+(\rho_k+\rho_B)(1-\bar{\rho})).$$

Similarly,

$$\mathcal{C}(I^*_{T\times N})-\mathcal{C}(I^2_{T\times N})$$
$$=\rho_k\mathbb{E}[\mathcal{R}]\left[\frac{\Delta_2}{(1-\bar{\rho})(1-\bar{\rho}-\rho_A-\rho_i)(1-\bar{\rho}-\rho_A-\rho_i-\rho_k)(1-\bar{\rho}-\rho_A-\rho_i-\rho_k-\rho_B)}\right],$$

where

$$\Delta_2 := -(c_A\mu_A\rho_A+c_i\mu_i\rho_i)(1-\bar{\rho}-\rho_A-\rho_i-\rho_k-\rho_B)$$
$$-c_B\mu_B\rho_B(1-\bar{\rho})+c_k\mu_k(\rho_B(1-\bar{\rho})+(\rho_A+\rho_i)(1-\bar{\rho}-\rho_A-\rho_i-\rho_k-\rho_B)).$$

The utilization for each type $l\in\mathcal{T}$ satisfies $0\le\rho_l<1$ and the total utilization (across arrivals for all types) is less than 1, therefore $0<1-\bar{\rho}-\rho_A-\rho_i-\rho_k-\rho_B\le 1-\bar{\rho}-\rho_A-\rho_i-\rho_k<1-\bar{\rho}-\rho_A-\rho_i<1-\bar{\rho}-\rho_A\le 1-\bar{\rho}$. The signs (positive or negative) of $\mathcal{C}(I^*_{T\times N})-\mathcal{C}(I^1_{T\times N})$ and $\mathcal{C}(I^*_{T\times N})-\mathcal{C}(I^2_{T\times N})$ depend only on the signs of Δ_1 and Δ_2, respectively. Further, recalling that $c_k\mu_k>c_i\mu_i$:

$$\Delta_1+\Delta_2=(c_k\mu_k-c_i\mu_i)\Big[(\rho_k+\rho_B)(1-\bar{\rho})+(\rho_A+\rho_i)(1-\bar{\rho}-\rho_A-\rho_i-\rho_k-\rho_B)\Big]>0.$$

so we conclude that at either $\Delta_1>0$ or $\Delta_2>0$ or both, implying that either $\mathcal{C}(I^*_{T\times N})-\mathcal{C}(I^1_{T\times N})>0$, or $\mathcal{C}(I^*_{T\times N})-\mathcal{C}(I^2_{T\times N})>0$ contradicting the optimality of $I^*_{T\times N}$. Q.E.D.

Proof of Proposition 3: The type-classifier-based classification probabilities are such that for all $i\in\mathcal{T}, j\in\mathcal{N}$, and β_t,

$$\widetilde{P}^t_{ij}(\beta_t)=\frac{\int_{\mathcal{X}} t_i(X)\,q^t_j(\beta_t,t(X))\,f(X)\,dX}{\int_{\mathcal{X}} t_i(X)f(X)\,dX}.$$

These match the theoretical classification probabilities

$$P^t_{ij}(\beta_t)=\int_{\mathcal{X}} q^t_j(\beta_t,t(X))f_i(X)\,dX,$$

for all $i\in\mathcal{T}, j\in\mathcal{N}$, and β_t if and only if

$$\int_{\mathcal{X}} q^t_j(\beta_t,t(X))\cdot h_i(X)dX=0,$$

for all $i\in\mathcal{T}, j\in\mathcal{N}$, and β_t, where

$$h_i(X)=\left[\frac{t_i(X)f(X)}{\int t_i(X)f(X)\,dX}-f_i(X)\right]$$

Therefore, as per the assumptions, $h(X)=0$ almost everywhere, i.e.,

$$\frac{t_i(X)f(X)}{\int t_i(X)f(X)\,dX}=f_i(X) \quad \text{almost everywhere.}$$

Further, let $A_i=\int_{\mathcal{X}} t_i(X)f(X)\,dX$. Then,

$$t_i(X)=\frac{A_i f_i(X)}{f(X)}.$$

Since $\sum_{i=1}^{T} t_i(X)=1$ for all X, we must have

$$\sum_{i=1}^{T} A_i f_i(X)=f(X)=\sum_{i=1}^{T} p_i f_i(X), \quad \forall X.$$

Now if $f_i(X)$ are linearly independent, then the only possibility is $A_i=p_i$ for all $i\in\mathcal{T}$ and $t_i(X)=\mathbb{P}(i|X)$. Q.E.D.

Proof of Lemma 1: For $T=N=2$, the value of average waiting cost $\mathcal{C}(P)$ (derived from equations 2 and 3) equals

$$\begin{aligned}\mathcal{C}(P)&=\mathbb{E}[\mathcal{R}]\Big[\frac{c_1\mu_1\rho_1P_{11}+c_2\mu_2\rho_2P_{21}}{1-\rho_1P_{11}-\rho_2P_{21}}+\frac{c_1\mu_1\rho_1(1-P_{11})+c_2\mu_2\rho_2(1-P_{21})}{(1-\rho_1P_{11}-\rho_2P_{21})(1-\rho)}\Big]\\&=\frac{(c_1\mu_1\rho_1+c_2\mu_2\rho_2)\mathbb{E}[\mathcal{R}]}{1-\rho}-\frac{(c_1\mu_1-c_2\mu_2)\mathbb{E}[\mathcal{R}]\rho_1\rho_2}{1-\rho}\cdot\frac{P_{11}-P_{21}}{1-\rho_1P_{11}-\rho_2P_{21}}\\&=\frac{(c_1\mu_1\rho_1+c_2\mu_2\rho_2)\,\mathbb{E}[\mathcal{R}]}{1-\rho}-\left(\frac{(c_1\mu_1-c_2\mu_2)\,\mathbb{E}[\mathcal{R}]\rho_1\rho_2}{1-\rho}\right)\cdot\frac{1-\Psi(P)}{\rho_2+\rho_1\Psi(P)}\end{aligned} \tag{36}$$

where

$$\Psi(P)=\frac{1-\rho P_{11}}{1-\rho P_{21}}. \tag{37}$$

Let $\widehat{P}(\tau)$ denote the type-classifier-based classification matrix for a prioritization policy that maps signal $z\geq\tau$ to queue 1 and to queue 2 otherwise. Then, $\widehat{P}_{11}(\tau)=\frac{\int_\tau^1 zb(z)dz}{\int_0^1 zb(z)dz}$ which equals $p_1^{-1}\int_\tau^1 zb(z)dz$ and $\widehat{P}_{21}(\tau)=p_2^{-1}\int_\tau^1(1-z)b(z)dz$ (by our assumed mean preservation). Since $\mathcal{C}(P)$ is an increasing function of $\Psi(P)$, the optimal threshold $\bar{\tau}$ minimizes $\phi_1(\tau)=\Psi(\widehat{P}(\tau))$:

$$\phi_1(\tau)=\frac{1-\rho\widehat{P}_{11}(\tau)}{1-\rho\widehat{P}_{21}(\tau)}=\frac{p_2}{p_1}\frac{(p_1-\rho\int_\tau^1 zb(z)dz)}{(p_2-\rho\int_\tau^1(1-z)b(z)dz)}$$

The derivative of $\phi_1(\tau)$ with respect to τ satisfies

$$\frac{d\phi_1(\tau)}{d\tau}=\frac{b(\tau)\rho p_2}{p_1}\frac{(\tau(p_2-\rho\int_\tau^1(1-z)b(z)dz)-(1-\tau)(p_1-\rho\int_\tau^1 zb(z)dz))}{(p_2-\rho\int_\tau^1(1-z)b(z)dz)^2}.$$

Since we assume $b(z)>0\ \forall z\in[0,1]$, the optimal τ satisfies

$$\eta(\tau):=\tau(p_2-\rho\int_\tau^1(1-z)b(z)dz)-(1-\tau)(p_1-\rho\int_\tau^1 zb(z)dz)=0$$

Now, $\eta(0) = -p_1(1-\rho) < 0$ and $\eta(1) = p_2 > 0$, and $\eta'(\tau)$ equals $(1-\rho\int_\tau^1 b(z)dz) > 0$. Therefore $\eta(\tau) = 0$ has a unique solution in $(0,1)$. Since for any $\tau > 0$, $\int_\tau^1 (1-z)b(z)dz < p_2$ and $\rho < 1$, therefore the optimal τ, i.e., $\bar{\tau}$, satisfies

$$\frac{\bar{\tau}}{1-\bar{\tau}} = \frac{p_1 - \rho\int_{\bar{\tau}}^1 zb(z)dz}{p_2 - \rho\int_{\bar{\tau}}^1 (1-z)b(z)dz}.$$

Next, we characterize the optimal prioritization threshold that minimizes the theoretical value of the average waiting cost. Let $P^t(\tau)$ denote the theoretical classification matrix for a policy that maps signal $z \geq \tau$ to queue 1. Then, $P^t_{11}(\tau) = p_1^{-1}\int_\tau^1 (z-\epsilon(z))b(z)dz$ and $P^t_{21}(\tau) = p_2^{-1}\int_\tau^1 (1-z+\epsilon(z))b(z)dz$. Then the optimal threshold minimizes

$$\phi_2(\tau) = \frac{p_2}{p_1}\frac{p_1 - \rho\int_\tau^1 (z-\epsilon(z))b(z)dz}{(p_2 - \rho\int_\tau^1 (1-z+\epsilon(z))b(z)dz)}.$$

Now,

$$\frac{d\phi_2(\tau)}{d\tau} = b(\tau)\frac{p_2}{p_1}\frac{((\tau-\epsilon(\tau))(p_2 - \rho\int_\tau^1 (1-z+\epsilon(z))b(z)dz) - (1-\tau+\epsilon(\tau))(p_1 - \rho\int_\tau^1 (z-\epsilon(z))b(z)dz))}{(p_2 - \rho\int_\tau^1 (1-z+\epsilon(z))b(z)dz)^2}.$$

Defining the numerator

$$\zeta(\tau) = ((\tau-\epsilon(\tau))(p_2 - \rho\int_\tau^1 (1-z+\epsilon(z))b(z))dz) - (1-\tau+\epsilon(\tau))(p_1 - \rho\int_\tau^1 (z-\epsilon(z))b(z)dz)),$$

we get $\zeta(0) < 0$, $\zeta(1) > 0$ (because $\tau - \epsilon(\tau)$ is 0 at $\tau = 0$ and 1 at $\tau = 1$ under the assumptions), and

$$\zeta'(\tau) = (1-\epsilon'(\tau))(1-\rho\int_\tau^1 b(z)dz).$$

Since $z - \epsilon(z)$ is an increasing function of z, $(1-\epsilon'(\tau)) > 0$, we have $\zeta'(\tau) > 0$. Therefore, there is a unique solution to $\frac{d\phi_2(\tau)}{d\tau} = 0$ in $(0,1)$, which is the optimal threshold τ^*:

$$\frac{\tau^* - \epsilon(\tau^*)}{1-\tau^*+\epsilon(\tau^*)} = \frac{p_1 - \rho\int_{\tau^*}^1 (z-\epsilon(z))b(z)dz}{p_2 - \rho\int_{\tau^*}^1 (1-z+\epsilon(z))b(z)}. \tag{38}$$

Further, since τ^* minimizes $\phi_2(\tau)$, we have $\phi_2(\tau^*) \leq \phi_2(\bar{\tau})$, and

$$\frac{\tau^* - \epsilon(\tau^*)}{1-(\tau^*-\epsilon(\tau^*))} = \frac{(p_1 - \rho\int_{\tau^*}^1 (z-\epsilon(z))b(z)dz)}{(p_2 - \rho\int_{\tau^*}^1 (1-z+\epsilon(z))b(z)dz)} \leq \frac{(p_1 - \rho\int_{\bar{\tau}}^1 (z-\epsilon(z))b(z)dz)}{(p_2 - \rho\int_{\bar{\tau}}^1 (1-z+\epsilon(z))b(z)dz)} \tag{39}$$

Under the assumptions of Lemma 1, $\epsilon(\tau) \leq 0$ for all $\tau \geq \bar{\tau}$. Therefore,

$$\frac{(p_1 - \rho\int_{\bar{\tau}}^1 zb(z)dz + \rho\int_{\bar{\tau}}^1 \epsilon(z)b(z)dz)}{(p_2 - \rho\int_{\bar{\tau}}^1 (1-z)b(z)dz - \rho\int_{\bar{\tau}}^1 \epsilon(z)b(z)dz)} \leq \frac{(p_1 - \rho\int_{\bar{\tau}}^1 zb(z)dz)}{(p_2 - \rho\int_{\bar{\tau}}^1 (1-z)b(z)dz)} = \frac{\bar{\tau}}{1-\bar{\tau}} \tag{40}$$

Since $\frac{x}{1-x}$ is increasing in x, and $\epsilon(\bar{\tau}) \leq 0$, we have

$$\frac{\bar{\tau}}{1-\bar{\tau}} \leq \frac{\bar{\tau}-\epsilon(\bar{\tau})}{1-\bar{\tau}+\epsilon(\bar{\tau})} \tag{41}$$

From (39) and (41), we have $\tau^* - \epsilon(\tau^*) \leq \bar{\tau} - \epsilon(\bar{\tau})$. Since, $z - \epsilon(z)$ is increasing in z, we have $\tau^* \leq \bar{\tau}$. Therefore, the probabilities that types 1 and 2 get classified into queue 1, i.e., $\int_{\bar{\tau}}^{1} (z - \epsilon(z))b(z)dz$ and $\int_{\bar{\tau}}^{1} (1 - z + \epsilon(z))b(z)dz$ are smaller than the optimal ($\int_{\tau^*}^{1} (z - \epsilon(z))b(z)dz$ and $\int_{\tau^*}^{1} (1 - z + \epsilon(z))b(z)dz$). Q.E.D.

Proof of Proposition 4: As in Lemma 1,

$$\arg\min_P \mathcal{C}(P) = \arg\min_P \frac{1 - \rho P_{11}}{1 - \rho P_{21}} = \arg\max_P \frac{\rho P_{11} - 1}{1 - \rho P_{21}},$$

and the optimization problem (21) can be reformulated as follows:

$$\max_P \frac{\rho P_{11} - 1}{1 - \rho P_{21}}$$

subject to:

$$TV_{12} - P_{11} + P_{21} \geq 0$$

$$TV_{12} - P_{21} + P_{11} \geq 0$$

$$1 - P_{11} \geq 0$$

$$1 - P_{21} \geq 0$$

$$P_{11} \geq 0$$

$$P_{21} \geq 0$$

Now, the first constraint $P_{21} \geq P_{11} - TV_{12}$ implies that

$$\frac{1}{1 - \rho P_{21}} \geq \frac{1}{1 - \rho P_{11} + \rho TV_{12}}.$$

Since $\rho P_{11} - 1 \leq 0$,

$$\frac{\rho P_{11} - 1}{1 - \rho P_{21}} \leq \frac{\rho P_{11} - 1}{1 - \rho P_{11} + \rho TV_{12}},$$

and

$$\max_P \frac{\rho P_{11} - 1}{1 - \rho P_{21}} \leq \max_P \frac{\rho P_{11} - 1}{1 - \rho P_{11} + \rho TV_{12}}$$

Further,

$$\frac{d}{dP_{11}} \left(\frac{\rho P_{11} - 1}{1 - \rho P_{11} + \rho TV_{12}} \right) = \frac{\rho(1 - \rho P_{11} + \rho TV_{12}) + \rho(\rho P_{11} - 1)}{(1 - \rho P_{11} + \rho TV_{12})^2} > 0.$$

Therefore, a maximum occurs at the greatest possible value of P_{11} which is 1 and $P_{21} = 1 - TV_{12}$. Since this solution satisfies all the constraints, it is the optimal solution to the original problem. Here, $TV_{12} = 0$ would mean that the types are indistinguishable from features and the waiting cost

is constant under all classification matrices. Q.E.D.

Proof of Proposition 5: Under the constraint $p_1P_{11}+p_2P_{21}=p_1$,

$$\frac{\rho P_{11}-1}{1-\rho P_{21}}=\frac{\rho P_{11}-1}{p_2-\rho p_1+\rho p_1P_{11}},$$

and on substituting $P_{21}=\frac{p_1(1-P_{11})}{p_2}$ in all the constraints, we get that the optimal P_{11} solves

$$\max \frac{\rho P_{11}-1}{p_2-\rho p_1+\rho p_1P_{11}}$$

subject to:

$$p_2TV_{12}+p_1-P_{11}\geq 0$$

$$p_2TV_{12}-p_1+P_{11}\geq 0$$

$$1-P_{11}\geq 0$$

$$p_2-p_1+p_1P_{11}\geq 0$$

$$P_{11}\geq 0$$

Now,

$$\frac{d}{dP_{11}}\left(\frac{\rho P_{11}-1}{p_2-\rho p_1+\rho p_1P_{11}}\right)=\frac{\rho(p_2-\rho p_1+\rho p_1P_{11})-\rho p_1(\rho P_{11}-1)}{(p_2-\rho p_1+\rho p_1P_{11})^2}>0$$

Hence the maximum of the objective (and equivalently the minimum of the waiting cost) occurs at the greatest possible value of P_{11} under the constraints, i.e., $P_{11}=p_1+p_2TV_{12}$. This is because $P_{11}=1$ is not feasible and $P_{11}=\frac{(p_1-p_2)}{p_1}<p_1$. Q.E.D.

Proof of Proposition 6: From (3), the average waiting cost $\mathcal{C}(P)$ equals

$$\mathbb{E}[\mathcal{R}]\Big[\frac{c_1\mu_1\rho_1P_{11}+c_2\mu_2\rho_2P_{21}+c_3\mu_3\rho_3P_{31}}{(1-\rho_1P_{11}-\rho_2P_{21}-\rho_3P_{31})}+\frac{c_1\mu_1\rho_1(1-P_{11})+c_2\mu_2\rho_2(1-P_{21})+c_3\mu_3\rho_3(1-P_{31})}{(1-\rho_1P_{11}-\rho_2P_{21}-\rho_3P_{31})(1-\rho)}\Big].$$

On simplifying,

$$\mathcal{C}(P)=\mathbb{E}[\mathcal{R}]\frac{c_1\mu_1\rho_1(1-\rho P_{11})+c_2\mu_2\rho_2(1-\rho P_{21})+c_3\mu_3\rho_3(1-\rho P_{31})}{(1-\rho_1P_{11}-\rho_2P_{21}-\rho_3P_{31})(1-\rho)}$$

Let

$$P^1=\begin{bmatrix} a+x & 0\\ b+x & y-x\\ c+x & z-x\end{bmatrix},\ P^2=\begin{bmatrix} a & x\\ b & y\\ c & z\end{bmatrix}.$$

Then,

$$\mathcal{C}(P^1)-\mathcal{C}(P^2)=\mathbb{E}[\mathcal{R}]\frac{-\rho x(c_1\mu_1\rho_1+c_2\mu_2\rho_2+c_3\mu_3\rho_3)}{(1-\rho_1a-\rho_2b-\rho_3c-\rho x)(1-\rho_1a-\rho_2b-\rho_3c)(1-\rho)}$$

Therefore, given any P_2 with $P_{12} > 0$, and under our assumption $P_{12} \leq P_{22} \leq P_{32}$, we can lower the waiting cost by generating P_1 by taking $x = P_{12}$. Therefore, the optimal solution has $P_{11} = 1$ and $P_{21} = 0$, and the optimization problem simplifies to

$$\min_P \frac{c_1\mu_1\rho_1(1-\rho) + c_2\mu_2\rho_2(1-\rho P_{21}) + c_3\mu_3\rho_3(1-\rho P_{31})}{(1-\rho_1-\rho_2 P_{21}-\rho_3 P_{31})}$$

subject to:

1. $TV_{12} - 1 + P_{21} \geq 0$
2. $TV_{13} - 1 + P_{31} \geq 0$
3. $TV_{23} - P_{21} + P_{31} \geq 0$
4. $1 - P_{21} \geq 0$
5. $1 - P_{31} \geq 0$
6. $P_{21} - P_{31} \geq 0$

Assumption 1 (Total Variation Inequalities) *The Proposition assumes $TV_{12} \leq TV_{13}$ and $TV_{23} \leq TV_{13}$. Using the triangle inequality and the definition of TV_{ij}, we get $TV_{12} + TV_{23} \geq TV_{13}$.*

Since the objective is a linear fraction with a positive denominator, the above problem can be equivalently transformed to a linear program—using Charnes–Cooper linearization (Charnes and Cooper 1962)—whose optimum occurs at a vertex (extreme point) of the feasible region. Therefore, the potential candidates of the optimum are the extreme points of the feasible region obtained by pairwise intersections of the linear constraints:

1. Constraints 1 and 2: $P_{21} = 1 - TV_{12}$, $P_{31} = 1 - TV_{13}$.
2. Constraints 1 and 3: $P_{21} = 1 - TV_{12}$, $P_{31} = 1 - TV_{12} - TV_{23}$; feasible under (2) only if $TV_{12} + TV_{23} = TV_{13}$
3. Constraints 1 and 4: Possible only when $TV_{12} = 0$. In this case, $P_{11} = P_{21} = 1$ and
$$\frac{\partial}{\partial P_{31}} \frac{c_1\mu_1\rho_1(1-\rho) + c_2\mu_2\rho_2(1-\rho) + c_3\mu_3\rho_3(1-\rho P_{31})}{(1-\rho_1-\rho_2-\rho_3 P_{31})} = \frac{\rho_3(1-\rho)((c_1\mu_1 - c_3\mu_3)\rho_1 + (c_2\mu_2 - c_3\mu_3)\rho_2)}{(1-\rho_1-\rho_2-\rho_3 P_{31})^2} > 0.$$
Therefore, the minimum occurs at the lowest value of P_{31} under the constraints, i.e., $1 - TV_{23}$.
4. Constraints 1 and 5: Only feasible if $TV_{12} = 0$ and implies $P_{21} = P_{31} = 1$. This is FIFO (first in first out) and is strictly suboptimal to the other solutions listed above.
5. Constraints 1 and 6: $P_{21} = 1 - TV_{12}$, $P_{31} = 1 - TV_{12}$.
6. Constraints 2 and 3: $P_{21} = 1 - TV_{13} + TV_{23}$, $P_{31} = 1 - TV_{13}$.
7. Constraints 2 and 4: $P_{21} = 1$, $P_{31} = 1 - TV_{13}$. Under Assum 1, possible only if $TV_{13} = TV_{23}$.
8. Constraints 2 and 5: Only feasible if $TV_{13} = 0$ and implies $P_{21} = P_{31} = 1$. This is FIFO and is strictly suboptimal to the other solutions listed above.

9. Constraints 2 and 6: $P_{21} = 1 - TV_{13}$, $P_{31} = 1 - TV_{13}$. Under Assum 1, possible only if $TV_{13} = TV_{12}$.
10. Constraints 3 and 4: $P_{21} = 1$, $P_{31} = 1 - TV_{23}$.
11. Constraints 3 and 5: Only feasible if $TV_{23} = 0$ and implies $P_{21} = P_{31} = 1$. This is FIFO and is strictly suboptimal to the other solutions listed above.
12. Constraints 3 and 6: Substituting $P_{31} = P_{21}$,

$$\frac{\partial}{\partial P_{21}} \frac{c_1\mu_1\rho_1(1-\rho) + c_2\mu_2\rho_2(1-\rho P_{21}) + c_3\mu_3\rho_3(1-\rho P_{21})}{(1-\rho_1-\rho_2 P_{21}-\rho_3 P_{21})} = \frac{\rho_1(1-\rho)[(c_1\mu_1 - c_2\mu_2)\rho_2 + (c_1\mu_1 - c_3\mu_3)\rho_3]}{(1-\rho_1-\rho_2 P_{21}-\rho_3 P_{21})^2} > 0.$$

    Therefore, the minimum occurs at the lowest feasible value of P_{21}, i.e., $1 - TV_{12}$.
13. Constraints 4 and 5 or 4 and 6 or 5 and 6: $P_{21} = P_{31} = 1$. This is FIFO and is strictly suboptimal to the other solutions listed above.

Therefore, the four candidates for the optimal classification matrix are

$$U = \begin{bmatrix} 1 & 0 \\ 1-TV_{12} & TV_{12} \\ 1-TV_{13} & TV_{13} \end{bmatrix}, \; V = \begin{bmatrix} 1 & 0 \\ 1-TV_{13}+TV_{23} & TV_{13}-TV_{23} \\ 1-TV_{13} & TV_{13} \end{bmatrix}, \; R = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1-TV_{23} & TV_{23} \end{bmatrix}, \; M = \begin{bmatrix} 1 & 0 \\ 1-TV_{12} & TV_{12} \\ 1-TV_{12} & TV_{12} \end{bmatrix}.$$

However, differentiating the objective with respect to P_{31} gives

$$\frac{\partial}{\partial P_{31}} \frac{c_1\mu_1\rho_1(1-\rho) + c_2\mu_2\rho_2(1-\rho P_{21}) + c_3\mu_3\rho_3(1-\rho P_{31})}{(1-\rho_1-\rho_2 P_{21}-\rho_3 P_{31})} = \frac{\rho_3[(c_1\mu_1 - c_3\mu_3)\rho_1(1-\rho) + (c_2\mu_2 - c_3\mu_3)\rho_2(1-\rho P_{21})]}{(1-\rho_1-\rho_2 P_{21}-\rho_3 P_{31})^2} > 0.$$

Therefore, given any value of P_{21}, the objective is strictly increasing in P_{31}. Since $TV_{13} \geq TV_{12}$, the classification matrix M is always dominated by the matrix U and cannot be the optimal solution.

Next, we compare the average waiting costs under the three potential optimum: U, V, and R.

For any classification matrix P

$$\mathcal{C}(P^{\text{FIFO}}) - \mathcal{C}(P) = \mathbb{E}[\mathcal{R}] \, \frac{\rho_1\rho_2\Delta_1(P_{11}-P_{21}) + \rho_2\rho_3\Delta_2(P_{21}-P_{31}) + \rho_1\rho_3\Delta_3(P_{11}-P_{31})}{(1-\rho)\left(1-\rho_1 P_{11}-\rho_2 P_{21}-\rho_3 P_{31}\right)}.$$

where

$$\Delta_1 = c_1\mu_1 - c_2\mu_2, \qquad \Delta_2 = c_2\mu_2 - c_3\mu_3, \qquad \Delta_3 = c_1\mu_1 - c_3\mu_3,$$

and

$$P^{\text{FIFO}} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}.$$

Case $P_{11} = 1$.

$$\mathcal{C}(P^{\text{FIFO}}) - \mathcal{C}(P) = \mathbb{E}[\mathcal{R}] \, \frac{(\rho_1\Delta_1 - \rho_3\Delta_2)\rho_2 P_{22} + (\rho_2\Delta_2 + \rho_1\Delta_3)\rho_3 P_{32}}{(1-\rho)\left(1-\rho+\rho_2 P_{22}+\rho_3 P_{32}\right)}.$$

Comparison of P and Q with $P_{11} = Q_{11} = 1$.

$$\begin{aligned}\mathcal{C}(P) - \mathcal{C}(Q) \le 0 &\iff \big(\mathcal{C}(P^{\text{FIFO}}) - \mathcal{C}(P)\big) - \big(\mathcal{C}(P^{\text{FIFO}}) - \mathcal{C}(Q)\big) \ge 0 \\ &\iff (1-\rho)\,(\rho_1\Delta_1 - \rho_3\Delta_2)\,\rho_2\,(P_{22} - Q_{22}) \\ &\quad + (1-\rho)\,(\rho_2\Delta_2 + \rho_1\Delta_3)\,\rho_3\,(P_{32} - Q_{32}) \\ &\quad - \Delta_2\,\rho\,\rho_2\rho_3\,(P_{22}Q_{32} - Q_{22}P_{32}) \ \ge 0.\end{aligned}$$

Conditions for optimality.

$$\begin{aligned}\mathcal{C}(U) - \mathcal{C}(R) \le 0 &\iff (1-\rho)(\rho_1\Delta_1 - \rho_3\Delta_2)\rho_2\,TV_{12} \\ &\quad + (1-\rho)(\rho_2\Delta_2 + \rho_1\Delta_3)\rho_3\,(TV_{13} - TV_{23}) \\ &\quad - \Delta_2\,\rho\,\rho_2\rho_3\,(TV_{12}\,TV_{23}) \ \ge 0 \\ &\iff \gamma \ge \gamma_1 = \frac{\rho_3}{\rho_1}\left[\frac{\frac{\rho\rho_2 TV_{23}}{1-\rho} + \rho_2 - (\rho_1+\rho_2)\frac{TV_{13}-TV_{23}}{TV_{12}}}{\rho_2 + \rho_3\frac{TV_{13}-TV_{23}}{TV_{12}}}\right].\end{aligned}$$

$$\begin{aligned}\mathcal{C}(V) - \mathcal{C}(R) \le 0 &\iff (TV_{13} - TV_{23})\Big[(1-\rho)\rho_1(\rho_2\Delta_1 + \rho_3\Delta_3) - \Delta_2\,\rho\,\rho_2\rho_3\,TV_{23}\Big] \ge 0 \\ &\iff \gamma \ge \gamma_2 = \frac{\rho\rho_2\rho_3 TV_{23} - \rho_3\rho_1(1-\rho)}{(\rho_2+\rho_3)\rho_1(1-\rho)}.\end{aligned}$$

$$\begin{aligned}\mathcal{C}(U) - \mathcal{C}(V) \le 0 &\iff (TV_{12} + TV_{23} - TV_{13})\Big[(1-\rho)(\rho_1\Delta_1 - \rho_3\Delta_2)\rho_2 - \Delta_2\,\rho\,\rho_2\rho_3\,TV_{13}\Big] \ge 0 \\ &\iff \gamma \ge \gamma_3 = \frac{\rho_3\,(1-\rho+\rho\,TV_{13})}{\rho_1(1-\rho)}.\end{aligned}$$

Ordering of thresholds. Next, we establish that $\gamma_2 \le \gamma_1 \le \gamma_3$ under Assumption 1

Let $\alpha := \frac{\rho}{1-\rho} > 0$, $a := \frac{TV_{13}-TV_{23}}{TV_{12}} \in [0,1]$, and $b := TV_{23} \in [0, TV_{13}]$. Since $\rho_3/\rho_1 > 0$, it suffices to compare the bracketed terms.

(i) $\gamma_1 \ge \gamma_2$. This holds if and only if the below holds:

$$\frac{\alpha\rho_2 b + \rho_2 - (\rho_1+\rho_2)a}{\rho_2 + \rho_3 a} \ \ge\ \frac{\alpha\rho_2 b - \rho_1}{\rho_2 + \rho_3}.$$

Cross-multiplying (positive denominators) and simplifying,

$$(\alpha\rho_2 b + \rho_2 - (\rho_1+\rho_2)a)(\rho_2+\rho_3) - (\alpha\rho_2 b - \rho_1)(\rho_2 + \rho_3 a) = (1-a)\,\rho_2\big(\rho + \alpha\rho_3 b\big) \ \ge\ 0.$$

Thus $\gamma_1 \ge \gamma_2$, with equality iff $a = 1$ (i.e., $TV_{13} = TV_{12} + TV_{23}$).

(ii) $\gamma_3 \geq \gamma_1$. We must show

$$1 + \alpha TV_{13} \;\geq\; \frac{\alpha\rho_2 b + \rho_2 - (\rho_1 + \rho_2)a}{\rho_2 + \rho_3 a}.$$

Cross-multiplying and simplifying,

$$(\rho_2 + \rho_3 a)(1 + \alpha TV_{13}) - (\alpha\rho_2 b + \rho_2 - (\rho_1 + \rho_2)a) = a\,\rho \;+\; \alpha\rho_2\,(TV_{13} - b) \;+\; \alpha\rho_3 a\,TV_{13} \;\geq\; 0,$$

since $a \geq 0$, $TV_{13} - b \geq 0$, $\rho > 0$, and $\alpha > 0$. Equality iff $a = 0$ and $b = TV_{13}$ (i.e., $TV_{13} = TV_{23}$).

Combining (i) and (ii) yields $\gamma_2 \leq \gamma_1 \leq \gamma_3$ under Assumption 1.

Optimal Policy.

- For $\gamma > \gamma_3$, $\mathcal{C}(U) < \mathcal{C}(V) < \mathcal{C}(R)$, and U is optimal.
- For $\gamma_2 \leq \gamma \leq \gamma_3$, $\mathcal{C}(V) \leq \mathcal{C}(U)$ and $\mathcal{C}(V) \leq \mathcal{C}(R)$, and V is optimal.
- For $\gamma < \gamma_2$ (only if $\gamma_2 > 0$), $\mathcal{C}(R) < \mathcal{C}(V) < \mathcal{C}(U)$, and R is optimal.

Q.E.D.

Proof of Proposition 7

For a given value of $i \in \mathcal{T},\ j \in \mathcal{N}$, and parameters β_d, let

$$\widehat{Y}^1_{ij}(\beta_d) = \frac{\sum_{s=1}^{S} y_{is} q^d_j(\beta_d, X_s)}{S},\ \widehat{Y}^2_{ij}(\beta_d) = \frac{\sum_{s=1}^{S} y_{is}}{S}$$

It immediately follows from the Strong Law of Large Numbers (SLLN),

$$\widehat{Y}^1_{ij}(\beta_d) \xrightarrow{a.s.} \mathbb{E}[y_{is} q^d_j(\beta_d, X_s)],\ \widehat{Y}^2_{ij}(\beta_d) \xrightarrow{a.s.} \mathbb{E}[y_{is}]$$

Since $\widehat{P}^d_{ij}(\beta_d) = \frac{\widehat{Y}^1_{ij}(\beta_d)}{\widehat{Y}^2_{ij}(\beta_d)}$, a continuous function of $(\widehat{Y}^1_{ij}(\beta_d), \widehat{Y}^2_{ij}(\beta_d))$, and $\mathbb{E}[y_{is}] = p_i > 0$, from the Continuous Mapping Theorem (CMT),

$$\widehat{P}^d_{ij}(\beta_d) = \frac{\widehat{Y}^1_{ij}(\beta_d)}{\widehat{Y}^2_{ij}(\beta_d)} \xrightarrow{a.s.} \frac{\mathbb{E}[y_{is} q^d_j(\beta_d, X_s)]}{\mathbb{E}[y_{is}]}.$$

Further,

$$\mathbb{E}[y_{is} q^d_j(\beta_d, X_s)] = p_i \mathbb{E}[q^d_j(\beta_d, X_s) | y_{is} = 1] + (1 - p_i) \times 0 = p_i \int_{X \in \mathcal{X}} q^d_j(\beta_d, X) f_i(X) dX.$$

Therefore,

$$\frac{\mathbb{E}[y_{is} q^d_j(\beta_d, X_s)]}{\mathbb{E}[y_{is}]} = P^d_{ij}(\beta_d).$$

Since $\mathcal{C}(\widehat{P}^d(\beta_d))$ is a continuous rational function of the variables $\widehat{P}^d_{ij}(\beta_d)$, the almost sure convergence of $\mathcal{C}(\widehat{P}^d(\beta_d))$ follows from the CMT.

Next, given that the parameter space $\mathcal{B}_d$ is compact and finite-dimensional and $q^d_j(\beta_d, X)$ is continuous in β_d for each X, we can establish uniform convergence of the empirical classification probabilities $\widehat{P}^d_{ij}(\beta_d)$ to their theoretical values $P^d_{ij}(\beta_d)$.

Uniform SLLN. Under the standing assumptions (i.i.d. data, $p_i := \mathbb{P}(Y_i = 1) > 0$), with $\mathcal{B}_d$ compact and $\beta_d \mapsto q_j^d(\beta_d, X)$ continuous for each X, Theorem 2 in Jennrich (1969) implies that

$$\sup_{\beta_d \in \mathcal{B}_d} |\widehat{Y}_{ij}^1(\beta_d) - \mathbb{E}[y_{is}\, q_j^d(\beta_d, X)]| \xrightarrow{a.s.} 0.$$

Moreover, $\sup_{\beta_d \in \mathcal{B}_d} |\widehat{Y}_{ij}^2(\beta_d) - \mathbb{E}[y_{is}]\,| \xrightarrow{a.s.} 0$. Since $\widehat{P}_{ij}^d(\beta_d) = \frac{\widehat{Y}_{ij}^1(\beta_d)}{\widehat{Y}_{ij}^2(\beta_d)}$, by a uniform ratio/continuous-mapping argument,

$$\sup_{\beta_d \in \mathcal{B}_d} \left|\widehat{P}_{ij}^d(\beta_d) - P_{ij}^d(\beta_d)\right| \xrightarrow{a.s.} 0.$$

Because $\rho < 1$, the map $P \mapsto \mathbb{E}[W_j(P)]$ is continuous with denominators uniformly bounded away from 0 under $\rho < 1$; hence $\mathcal{C}(P)$ is uniformly continuous on the compact domain $(\Delta^N)^T$. Therefore, from uniform CMT

$$\sup_{\beta_d \in \mathcal{B}_d} \left|\mathcal{C}(\widehat{P}^d(\beta_d)) - \mathcal{C}(P^d(\beta_d))\right| \xrightarrow{a.s.} 0.$$

Next, we establish the convergence of the optimal empirical cost to the optimal theoretical cost. Let $(X_s, y_s)_{s \geq 1}$ be the underlying i.i.d. sequence of data, defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Each element $\omega \in \Omega$ corresponds to one infinite realization of the dataset $(X_1(\omega), y_1(\omega)), (X_2(\omega), y_2(\omega)), \ldots$.

From the uniform almost-sure convergence result established above, there exists a measurable subset $\Omega^\star \subseteq \Omega$ with $\mathbb{P}(\Omega^\star) = 1$ such that for every $\omega \in \Omega^\star$,

$$\lim_{S \to \infty} \sup_{\beta_d \in \mathcal{B}_d} \left|\mathcal{C}(\widehat{P}^d(\beta_d; \omega)) - \mathcal{C}(P^d(\beta_d))\right| = 0.$$

Here the notation $\widehat{P}^d(\beta_d; \omega)$ emphasizes that the empirical probabilities depend on the realized sample path ω.

Fix $\varepsilon > 0$ and $\omega \in \Omega^\star$. By the ε–N definition of the real limit, there exists an index $S_0 = S_0(\omega, \varepsilon)$ such that for all $S \geq S_0$,

$$\sup_{\beta_d \in \mathcal{B}_d} \left|\mathcal{C}(\widehat{P}^d(\beta_d; \omega)) - \mathcal{C}(P^d(\beta_d))\right| \leq \varepsilon.$$

This inequality implies that, for all $\beta_d \in \mathcal{B}_d$ and all $S \geq S_0$,

$$\mathcal{C}(P^d(\beta_d)) - \varepsilon \;\leq\; \mathcal{C}(\widehat{P}^d(\beta_d; \omega)) \;\leq\; \mathcal{C}(P^d(\beta_d)) + \varepsilon. \tag{42}$$

Now let $\beta_d^\star \in \arg\min_{\beta_d \in \mathcal{B}_d} \mathcal{C}(P^d(\beta_d))$. From the left inequality in (42),

$$\min_{\beta_d} \mathcal{C}(\widehat{P}^d(\beta_d; \omega)) \;\geq\; \min_{\beta_d} \mathcal{C}(P^d(\beta_d)) - \varepsilon.$$

From the right inequality in (42) evaluated at $\beta_d^\star$,

$$\min_{\beta_d} \mathcal{C}(\widehat{P}^d(\beta_d; \omega)) \;\leq\; \mathcal{C}(\widehat{P}^d(\beta_d^\star; \omega)) \;\leq\; \mathcal{C}(P^d(\beta_d^\star)) + \varepsilon = \min_{\beta_d} \mathcal{C}(P^d(\beta_d)) + \varepsilon.$$

Together these inequalities yield that for all $\omega \in \Omega^*$

$$\left| \min_{\beta_d} \mathcal{C}(\widehat{P}^d(\beta_d;\omega)) - \min_{\beta_d} \mathcal{C}(P^d(\beta_d)) \right| \le \varepsilon \quad \text{for all } S \ge S_0,$$

so

$$\min_{\beta_d} \mathcal{C}(\widehat{P}^d(\beta_d)) \xrightarrow{a.s.} \min_{\beta_d} \mathcal{C}(P^d(\beta_d)).$$

Finally, we establish the convergence of the empirical minimizer's cost to the optimal theoretical cost. Let $\widehat{\beta}_d \in \arg\min_{\beta_d} \mathcal{C}(\widehat{P}^d(\beta_d;\omega))$. Applying (42) to $\widehat{\beta}_d$ and then to $\beta_d^\star$, we obtain

$$\begin{aligned}\mathcal{C}(P^d(\widehat{\beta}_d)) &\le \mathcal{C}(\widehat{P}^d(\widehat{\beta}_d;\omega)) + \varepsilon = \min_{\beta_d} \mathcal{C}(\widehat{P}^d(\beta_d;\omega)) + \varepsilon \\ &\le \mathcal{C}(\widehat{P}^d(\beta_d^\star;\omega)) + \varepsilon \le \mathcal{C}(P^d(\beta_d^\star)) + 2\varepsilon = \min_{\beta_d} \mathcal{C}(P^d(\beta_d)) + 2\varepsilon.\end{aligned}$$

Since $\min_{\beta_d} \mathcal{C}(P^d(\beta_d)) \le \mathcal{C}(P^d(\widehat{\beta}_d))$, it follows that

$$0 \le \mathcal{C}(P^d(\widehat{\beta}_d)) - \min_{\beta_d} \mathcal{C}(P^d(\beta_d)) \le 2\varepsilon \quad \text{for all } S \ge S_0.$$

Because $\varepsilon > 0$ is arbitrary and the above result holds for all $\omega \in \Omega^*$, we conclude that

$$\mathcal{C}(P^d(\widehat{\beta}_d)) \xrightarrow{a.s.} \min_{\beta_d} \mathcal{C}(P^d(\beta_d)).$$

Q.E.D.

Proof of Proposition 8

Using the chain rule of differentiation for each entry β_{dk} of β_d, where $k \in \mathcal{D}$,

$$\frac{\partial \mathcal{C}(\widehat{P}^d(\beta_d))}{\partial \beta_{dk}} = \sum_{i=1}^{T} \sum_{j=1}^{N} \frac{\partial \mathcal{C}(\widehat{P}^d(\beta_d))}{\partial \widehat{P}_{ij}^d(\beta_d)} \times \frac{\partial \widehat{P}_{ij}^d(\beta_d)}{\partial \beta_{dk}}.$$

Since $\frac{\partial \mathcal{C}(\widehat{P}^d(\beta_d))}{\partial \beta_{dk}}$ is a continuous function of the variables $\frac{\partial \mathcal{C}(\widehat{P}^d(\beta_d))}{\partial \widehat{P}_{ij}^d(\beta_d)}$ and $\frac{\partial \widehat{P}_{ij}^d(\beta_d)}{\partial \beta_{dk}}$, $i \in \mathcal{T},\ j \in \mathcal{N}$, it is sufficient to prove that each of these estimates converges almost surely to its expected value.

The value of $\mathcal{C}(P)$ (defined in equation (3) in §2) equals

$$\mathcal{C}(P) = \sum_{l \in \mathcal{N}} \mathbb{E}[W_l(P)] (\sum_{k \in \mathcal{T}} c_k \mu_k \rho_k P_{kl}).$$

Therefore,

$$\frac{\partial \mathcal{C}(P)}{\partial P_{ij}} = c_i \mu_i \rho_i \mathbb{E}[W_j(P)] + \sum_{l \in \mathcal{N}} \frac{\partial \mathbb{E}[W_l(P)]}{\partial P_{ij}} (\sum_{k \in \mathcal{T}} c_k \mu_k \rho_k P_{kl}),$$

where, recall,

$$\mathbb{E}[W_l(P)] = \frac{\mathbb{E}[\mathcal{R}]}{(1 - \sum_{r<l} \sum_{m \in \mathcal{T}} \rho_m P_{mr}) \left(1 - \sum_{r \le l} \sum_{m \in \mathcal{T}} \rho_m P_{mr}\right)}.$$

On substituting the values of $\mathbb{E}[W_j(P)]$ and $\frac{\partial \mathbb{E}[W_l(P)]}{\partial P_{ij}}$, we get

$$\frac{\partial \mathcal{C}(P)}{\partial P_{ij}} = c_i \mu_i \rho_i \cdot \frac{\mathbb{E}[\mathcal{R}]}{\left(1-\sum_{r<j}\sum_{m\in\mathcal{T}}\rho_m P_{mr}\right)\left(1-\sum_{r\le j}\sum_{m\in\mathcal{T}}\rho_m P_{mr}\right)}$$
$$+\sum_{l\ge j}\left(\sum_{k\in\mathcal{T}} c_k\mu_k\rho_k P_{kl}\right)\cdot \rho_i \cdot \mathbb{E}[\mathcal{R}]\cdot\left[\frac{\mathbb{I}_{j<l}}{\left(1-\sum_{r<l}\sum_{m\in\mathcal{T}}\rho_m P_{mr}\right)^2\left(1-\sum_{r\le l}\sum_{m\in\mathcal{T}}\rho_m P_{mr}\right)}\right.$$
$$\left.+\frac{\mathbb{I}_{j\le l}}{\left(1-\sum_{r<l}\sum_{m\in\mathcal{T}}\rho_m P_{mr}\right)\left(1-\sum_{r\le l}\sum_{m\in\mathcal{T}}\rho_m P_{mr}\right)^2}\right]$$

Therefore, for each $i\in\mathcal{T},\ j\in\mathcal{N},\ \frac{\partial\mathcal{C}(\widehat{P}^d(\beta_d))}{\partial\widehat{P}^d_{ij}(\beta_d)}$ is a continuous rational function of the variables $\widehat{P}^d_{kl}(\beta_d),\ k\in\mathcal{T},\ l\in\mathcal{N}$. Since each of the variables $\widehat{P}^d_{kl}(\beta_d)\xrightarrow{a.s.}P^d_{kl}(\beta_d)$ (from Proposition 7), from the CMT, for all $i\in\mathcal{T}$, $j\in\mathcal{N}$, and β_d,

$$\frac{\partial\mathcal{C}(\widehat{P}^d(\beta_d))}{\partial\widehat{P}^d_{ij}(\beta_d)}\xrightarrow{a.s.}\frac{\partial\mathcal{C}(P^d(\beta_d))}{\partial P^d_{ij}(\beta_d)}.$$

Further,

$$\frac{\partial\widehat{P}^d_{ij}(\beta_d)}{\partial\beta_{dk}}=\frac{\sum_{s=1}^S y_{is}\frac{\partial q_j(\beta_d,X_s)}{\partial\beta_{dk}}}{\sum_{s=1}^S y_{is}}$$

If $\mathbb{E}[|y_{is}\frac{\partial q_j(\beta_d,X_s)}{\partial\beta_{dk}}|]=p_i\int_{X\in\mathcal{X}}\left|\frac{\partial q_j(\beta_d,X)}{\partial\beta_{dk}}\right|f_i(X)dX<\infty$, as assumed, then from the SLLN

$$\frac{\partial\widehat{P}^d_{ij}(\beta_d)}{\partial\beta_{dk}}\xrightarrow{a.s.}\int_{X\in\mathcal{X}}\frac{\partial q_j(\beta_d,X)}{\partial\beta_{dk}}f_i(X)dX=\frac{\partial}{\partial\beta_{dk}}\int_{X\in\mathcal{X}}q_j(\beta_d,X)f_i(X)dX=\frac{\partial P^d_{ij}(\beta_d)}{\partial\beta_{dk}}$$

Thus, for each fixed β_d and all $i\in\mathcal{T}$, $j\in\mathcal{N}$, $k\le D$,

$$\frac{\partial\mathcal{C}(\widehat{P}^d(\beta_d))}{\partial\widehat{P}^d_{ij}(\beta_d)}\xrightarrow{a.s.}\frac{\partial\mathcal{C}(P^d(\beta_d))}{\partial P^d_{ij}(\beta_d)}\quad\text{and}\quad\frac{\partial\widehat{P}^d_{ij}(\beta_d)}{\partial\beta_{dk}}\xrightarrow{a.s.}\frac{\partial P^d_{ij}(\beta_d)}{\partial\beta_{dk}}.$$

By the chain rule,

$$\nabla_{\beta_d}\mathcal{C}(\widehat{P}^d(\beta_d))\xrightarrow{a.s.}\nabla_{\beta_d}\mathcal{C}(P^d(\beta_d)),$$

establishing pointwise gradient convergence.

Uniform gradient convergence. Given that $\mathcal{B}_d$ is compact and, for each i,j,k, the class

$$\left\{(X,y)\mapsto y_i\frac{\partial q_j(\beta_d,X)}{\partial\beta_{dk}}:\ \beta_d\in\mathcal{B}_d\right\}$$

is dominated (because $\frac{\partial q_j(\beta_d,X)}{\partial\beta_{dk}}$ is uniformly bounded and there exists $M_1<\infty$ for which $|\frac{\partial q_j(\beta_d,X)}{\partial\beta_{dk}}|\le M_1$), measurable, and continuous in β_d on $\mathcal{B}_d$, Theorem 2 in Jennrich (1969) implies a uniform SLLN on compact parameter sets, yielding

$$\sup_{\beta_d\in\mathcal{B}_d}\left|\frac{1}{S}\sum_{s=1}^S y_{is}\frac{\partial q^d_j(\beta_d,X_s)}{\partial\beta_{dk}}-\mathbb{E}[y_{is}\frac{\partial q^d_j(\beta_d,X_s)}{\partial\beta_{dk}}]\right|\xrightarrow{a.s.}0,$$

and with $\frac{1}{S}\sum_s y_{is} \to p_i > 0$ a.s. the uniform ratio gives

$$\sup_{\beta_d \in \mathcal{B}_d} \max_{i\in\mathcal{T},\ j\in\mathcal{N},\ k\in\{1,\dots,D\}} \left| \frac{\partial \widehat{P}^d_{ij}(\beta_d)}{\partial \beta_{dk}} - \frac{\partial P^d_{ij}(\beta_d)}{\partial \beta_{dk}} \right| \xrightarrow{a.s.} 0.$$

Together with $\sup_{\beta_d} \|\widehat{P}^d(\beta_d) - P^d(\beta_d)\| \to 0$ a.s. and continuity of $P \mapsto \frac{\partial \mathcal{C}(P)}{\partial P_{ij}}$, the chain rule and CMT imply

$$\sup_{\beta_d \in \mathcal{B}_d} \|\nabla_{\beta_d}\mathcal{C}(\widehat{P}^d(\beta_d)) - \nabla_{\beta_d}\mathcal{C}(P^d(\beta_d))\| \xrightarrow{a.s.} 0.$$

Finally, let $\widehat{\beta}_d \in \arg\min_{\beta_d} \mathcal{C}(\widehat{P}^d(\beta_d))$ be an empirical stationary point, so $\nabla_{\beta_d}\mathcal{C}(\widehat{P}^d(\widehat{\beta}_d)) = 0$. Then

$$\|\nabla_{\beta_d}\mathcal{C}(P^d(\widehat{\beta}_d))\| \le \sup_{\beta_d \in \mathcal{B}_d} \|\nabla_{\beta_d}\mathcal{C}(P^d(\beta_d)) - \nabla_{\beta_d}\mathcal{C}(\widehat{P}^d(\beta_d))\| + \|\nabla_{\beta_d}\mathcal{C}(\widehat{P}^d(\widehat{\beta}_d))\|.$$

By the uniform gradient convergence just established, the supremum term converges to 0 almost surely, while the second term equals 0 by stationarity. Hence $\nabla_{\beta_d}\mathcal{C}(P^d(\widehat{\beta}_d)) \xrightarrow{a.s.} 0$. Q.E.D.

Proof of Lemma 2:

As shown in Appendix §7.4, the value $\mathcal{C}(\widehat{P}(\beta_d))$ equals

$$\mathcal{C}(\widehat{P}(\beta_d)) = \frac{\mathbb{E}[\mathcal{R}]}{(1-\rho)}\Big[\frac{K - \rho\sum_{s=1}^S \eta_s q_1^d(\beta_d, X_s)}{(1-\sum_{s=1}^S \gamma_s q_1^d(\beta_d, X_s))}\Big]$$

where

$$K = \sum_{i=1}^T c_i\mu_i\rho_i,\ S_i = \sum_{s=1}^S y_{is},\ \eta_s = \sum_{i=1}^T \frac{c_i\mu_i\rho_i y_{is}}{S_i},\ \gamma_s = \sum_{i=1}^T \frac{\rho_i y_{is}}{S_i},\ \mathbb{E}[\mathcal{R}] = \sum_{i=1}^T \frac{\lambda_i}{\mu_i^2}.$$

Since, the first and second partial derivatives are bounded, there exist constants $M_1, M_2 > 0$ such that

$$\left|\frac{\partial q_1^d(\beta_d, X)}{\partial \beta_{d,k}}\right| \le M_1, \quad \left|\frac{\partial^2 q_1^d(\beta_d, X)}{\partial \beta_{d,k}\partial \beta_{d,l}}\right| \le M_2, \quad \forall k,l \in \{1,\dots,D\},\ X \in \mathbb{R}^p.$$

Next, we will establish that the first and second partial derivatives of the empirical waiting cost $\mathcal{C}(\widehat{P}(\beta_d))$ are bounded, and the gradient of the empirical waiting cost $\nabla_{\beta_d}\mathcal{C}(\widehat{P}(\beta_d))$ is Lipschitz continuous, which implies the convergence of gradient descent.

$$\frac{\partial \mathcal{C}(\widehat{P}(\beta_d))}{\partial \beta_{di}} = K\cdot\mathbb{E}[\mathcal{R}]\Big[\frac{(1-\rho\sum_{s=1}^S \frac{\eta_s}{K} q_1^d(\beta_d, X_s))(\sum_{s=1}^S \gamma_s \frac{\partial q_1^d(\beta_d, X_s)}{\partial \beta_{di}}) - (1-\sum_{s=1}^S \gamma_s q_1^d(\beta_d, X_s))(\rho\sum_{s=1}^S \frac{\eta_s}{K}\frac{\partial q_1^d(\beta_d, X_s)}{\partial \beta_{di}})}{(1-\sum_{s=1}^S \gamma_s q_1^d(\beta_d, X_s))^2(1-\rho)}\Big].$$

On using the triangle inequality,

$$\left|\frac{\partial \mathcal{C}(\widehat{P}(\beta_d))}{\partial \beta_{di}}\right| \le K\cdot\mathbb{E}[\mathcal{R}]\cdot\frac{\left[\sum_{s=1}^S \gamma_s \left|\frac{\partial q_1^d(\beta_d, X_s)}{\partial \beta_{di}}\right|\right]\cdot\left|1-\rho\sum_{s=1}^S \frac{\eta_s}{K} q_1^d(\beta_d, X_s)\right|}{\left(1-\sum_{s=1}^S \gamma_s q_1^d(\beta_d, X_s)\right)^2(1-\rho)}$$
$$+ K\cdot\mathbb{E}[\mathcal{R}]\cdot\frac{\left|1-\sum_{s=1}^S \gamma_s q_1^d(\beta_d, X_s)\right|\cdot\left[\rho\sum_{s=1}^S \frac{\eta_s}{K}\left|\frac{\partial q_1^d(\beta_d, X_s)}{\partial \beta_{di}}\right|\right]}{\left(1-\sum_{s=1}^S \gamma_s q_1^d(\beta_d, X_s)\right)^2(1-\rho)}$$

Further, $0 \leq q_1^d(\beta_d, X_s) \leq 1$, $\sum_{s=1}^S \gamma_s = \rho$, $\sum_{s=1}^S \frac{\eta_s}{K} = 1$, and

$$\left(1 - \sum_{s=1}^S \gamma_s q_1^d(\beta_d, X_s)\right)^2 \geq (1-\rho)^2,$$

$$\left|\frac{\partial q_1^d(\beta_d, X_s)}{\partial \beta_{di}}\right| \leq M_1, \quad \left|1 - \sum_{s=1}^S \gamma_s q_1^d(\beta_d, X_s)\right| < 1, \quad \left|1 - \rho \sum_{s=1}^S \frac{\eta_s}{K} q_1^d(\beta_d, X_s)\right| < 1.$$

Therefore,

$$\boxed{\left|\frac{\partial \mathcal{C}(\widehat{P}^d(\beta_d))}{\partial \beta_{di}}\right| \leq \frac{2\rho M_1}{(1-\rho)^3} \cdot K \cdot \mathbb{E}[\mathcal{R}]}$$

Further, on differentiating $\frac{\partial \mathcal{C}(\widehat{P}(\beta_d))}{\partial \beta_{di}}$ with respect to β_{dj}, we get

$$\frac{\partial^2 \mathcal{C}(\widehat{P}^d(\beta_d))}{\partial \beta_{di} \partial \beta_{dj}} = K \cdot \mathbb{E}[\mathcal{R}] \cdot \frac{(1 - \sum_{s=1}^S \gamma_s q_1^d(\beta_d, X_s))^2 \frac{\partial N(\beta_d)}{\partial \beta_{dj}} + 2N(\beta_d) \sum_{s=1}^S \gamma_s \frac{q_1^d(\beta_d, X_s)}{\partial \beta_{dj}} (1 - \sum_{s=1}^S \gamma_s q_1^d(\beta_d, X_s))}{(1 - \sum_{s=1}^S \gamma_s q_1^d(\beta_d, X_s))^4 (1-\rho)}$$

where

$$N(\beta_d) = (1 - \rho \sum_{s=1}^S \frac{\eta_s}{K} q_1^d(\beta_d, X_s))(\sum_{s=1}^S \gamma_s \frac{\partial q_1^d(\beta_d, X_s)}{\partial \beta_{di}}) - (1 - \sum_{s=1}^S \gamma_s q_1^d(\beta_d, X_s))(\rho \sum_{s=1}^S \frac{\eta_s}{K} \frac{\partial q_1^d(\beta_d, X_s)}{\partial \beta_{di}})$$

and

$$\begin{aligned}
\frac{\partial N(\beta_d)}{\partial \beta_{dj}} = &- \left(\rho \sum_{s=1}^S \frac{\eta_s}{K} \frac{\partial q_1^d(\beta_d, X_s)}{\partial \beta_{dj}}\right) \cdot \left(\sum_{s=1}^S \gamma_s \frac{\partial q_1^d(\beta_d, X_s)}{\partial \beta_{di}}\right) \\
&+ \left(1 - \rho \sum_{s=1}^S \frac{\eta_s}{K} q_1^d(\beta_d, X_s)\right) \cdot \left(\sum_{s=1}^S \gamma_s \frac{\partial^2 q_1^d(\beta_d, X_s)}{\partial \beta_{di} \partial \beta_{dj}}\right) \\
&+ \left(\sum_{s=1}^S \gamma_s \frac{\partial q_1^d(\beta_d, X_s)}{\partial \beta_{dj}}\right) \cdot \left(\rho \sum_{s=1}^S \frac{\eta_s}{K} \frac{\partial q_1^d(\beta_d, X_s)}{\partial \beta_{di}}\right) \\
&- \left(1 - \sum_{s=1}^S \gamma_s q_1^d(\beta_d, X_s)\right) \cdot \left(\rho \sum_{s=1}^S \frac{\eta_s}{K} \frac{\partial^2 q_1^d(\beta_d, X_s)}{\partial \beta_{di} \partial \beta_{dj}}\right)
\end{aligned}$$

Using the triangle inequality, we can show that

$$\left|\frac{\partial^2 \mathcal{C}(\widehat{P}^d(\beta_d))}{\partial \beta_{di} \partial \beta_{dj}}\right| \leq K \cdot \mathbb{E}[\mathcal{R}] \cdot \frac{\left|\frac{\partial N(\beta_d)}{\partial \beta_{dj}}\right| + 2\rho M_1 \left|N(\beta_d)\right|}{(1-\rho)^5}$$

and

$$|N(\beta_d)| \leq 2\rho M_1 \quad \text{and} \quad \left|\frac{\partial N(\beta_d)}{\partial \beta_{dj}}\right| \leq 2\rho^2 M_1^2 + 2\rho M_2.$$

Therefore,

$$\boxed{\left|\frac{\partial^2 \mathcal{C}(\widehat{P}^d(\beta_d))}{\partial \beta_{di} \partial \beta_{dj}}\right| \leq \frac{(6M_1^2 \rho^2 + 2M_2 \rho)}{(1-\rho)^5} \cdot K \cdot \mathbb{E}[\mathcal{R}]}$$

Therefore,

$$\sup_{\beta_d} \left\| \nabla^2 \mathcal{C}(\widehat{P}^d(\beta_d)) \right\|_2 = \sup_{\beta_d} \sqrt{\sum_{i=1}^{D} \sum_{j=1}^{D} \left| \frac{\partial^2 \mathcal{C}(\widehat{P}^d(\beta_d))}{\partial \beta_{di} \partial \beta_{dj}} \right|^2} \leq D \cdot \frac{(6M_1^2\rho^2 + 2M_2\rho)}{(1-\rho)^5} K \cdot \mathbb{E}[\mathcal{R}].$$

Further,

$$\left\| \nabla_{\beta_d} \mathcal{C}(\widehat{P}^d(\beta_d^a)) - \nabla_{\beta_d} \mathcal{C}(\widehat{P}^d(\beta_d^b)) \right\|_2 = \left\| \int_{\tau=0}^{1} \nabla^2_{\beta_d} \mathcal{C} \left(\widehat{P}^d(\beta_d^b + \tau(\beta_d^a - \beta_d^b)) \right) d\tau \cdot (\beta_d^b - \beta_d^a) \right\|_2$$

Using the triangle inequality for integrals

$$\begin{aligned} \left\| \int_{\tau=0}^{1} \nabla^2 \mathcal{C} \left(\widehat{P}^d \left(\beta_d^b + \tau(\beta_d^a - \beta_d^b) \right) \right) d\tau \cdot (\beta_d^b - \beta_d^a) \right\|_2 &\leq \left(\int_{\tau=0}^{1} \left\| \nabla^2_{\beta_d} \mathcal{C} \left(\widehat{P}^d \left(\beta_d^b + \tau(\beta_d^a - \beta_d^b) \right) \right) \right\|_2 d\tau \right) \cdot \left\| \beta_d^b - \beta_d^a \right\|_2 \\ &\leq L \left\| \beta_d^b - \beta_d^a \right\|_2 \end{aligned}$$

where

$$L = D \cdot \frac{(6M_1^2\rho^2 + 2M_2\rho)}{(1-\rho)^5} \cdot K \cdot \mathbb{E}[\mathcal{R}]$$

Therefore, from Proposition 3.3.1 and Lemma 3.3.1 in Nemirovski (2009),

$$\min_{0 \leq m < M} \left\| \nabla_{\beta_d} \mathcal{C}(\widehat{P}^d(\beta_d^m)) \right\|_2^2 \leq \frac{2L}{M} \Big(\mathcal{C}(\widehat{P}^d(\beta_d^0)) - \min_{\beta_d} \mathcal{C}(\widehat{P}^d(\beta_d)) \Big).$$

Q.E.D.

Extension to $N \geq 3$. We presented the details for $N = 2$ to keep the algebra tractable, but the same idea extends to any $N \geq 3$. The cost $\mathcal{C}(\widehat{P}^d(\beta_d))$ remains a *sum of rational functions* of the queue–assignment probabilities: the numerators are linear sums, and the denominators are the same load/utilization terms. Because the overall load satisfies $\rho < 1$, these denominators stay bounded away from zero, so the expression is well behaved. Adding more queues just adds more of the same rational pieces, which keeps the gradient and Hessian bounded for the same reason (the constants may grow with N, but are finite for fixed N). Consequently, the same gradient–descent convergence argument carries over, unchanged in spirit, to all $N \geq 3$.